0.1 Template C++

```
#pragma GCC optimize("O3")
const int OO=(int)1e9;
const ll INF=(ll)2e18;
#define FOR(i,n) for(int i=0;i<n;i++)
#define REPP(i,l,r,c) for(int i=l; i<=r; i+=c)
#define REP(i,l,r) REPP(i,l,r,1)
#define FORD(i,n) for(int i=n-1; i>=0; i--)
#define REVV(i,l,r,c) for(int i=l; i>=r; i-=c)
#define REV(i,l,r) REVV(i,l,r,1)

int irand(int lo,int hi){ // random
  return ((double)rand()/(RAND_MAX+1.0)) * \
    (hi-lo+1) + lo;
}
string toString(ll x){ // ll to string
  stringstream ss;  ss << x;  return ss.str();
}
ll toNumber(string S){ // string to ll
  ll ret;  sscanf(S.c_str(),"%lld",&ret);
    return ret;
}
void ioSync(){ //iostream sync
  ios_base::sync_with_stdio(0);
  cin.tie(0);  cout.tie(0);
}
```

0.2 Template Java

```
import java.math.BigInteger;
import java.util.Scanner;
import java.util.ArrayList;

class Main{ // must be named as Main
  static ArrayList A=new ArrayList<BigInteger>();
  static void init(){
    BigInteger x = new BigInteger("1656");
    BigInteger y = BigInteger.valueOf(123);
    x = x.add(new BigInteger("4498"));
    // substract  multiply  divide  remainder
    // compare -> A.compareTo(B) < 0  A.equals(B)
    int x = A.intValue();
    long y = A.longValue();
    String s = A.toString();
  }
  public static void Main(String[] args){
    init();  int x;
    Scanner in = new Scanner(System.in);
    while(in.hasNext()){
      System.out.println(">> "+in.NextInt());
} } }
```

0.3 Template Python 3.6

```
# lib: math cmath queue heapq collections.deque
class Data:
  def __init__(self, _name, _num):
    self.name, self.num = _name, _num;
  def __lt__(self, rhs): return self.num < rhs.num;
def main():
  temp = input().split();
  n,s,data = int(temp[0]), int(temp[1]), [];
  # input
  for i in range(n):
    temp = input().split();
    data.append( Data(temp[0], int(temp[1])) );
  # do
  data.sort(reverse=True);
  res = [];
  for i in range(n):
    if(s < data[i].num): continue;
    s -= data[i].num;
    res.append(i);
  if(s != 0): res = [];
  # print
  print(len(res));
  for x in res: print(data[x].name);
main();
```

0.4 Bit Operations

```
int removeLSOne(int num){   return num&(num-1); }
__builtin_popcount(x)// count 1-bit, (_popcountll)
__builtin_clz(x)// count of leading 0 (_clzll)
__builtin_ctz(x)// count of trailing 0 (_ctzll)
```

1 Data Structures

1.1 Efficient Segment Tree

```
// array size: 2N     data inside N..2N-1 0-based

void build(){ // build the tree
  REV(i,n-1,1)t[i] = t[i<<1]+t[i<<1|1];
}
void updPURQ(int pos, int val){
  for(t[pos+=n] = val; pos>>=1;)
    t[pos] = combine(t[pos<<1], t[pos<<1|1]);
}
int queryPURQ(int l, int r){
  int resa = 0, resb = 0;
  for(l+=n,r+=n; l<r; l>>=1,r>>=1){
    if(l&1)res = combine(resa, t[l++]);
    if(r&1)res = combine(t[--r], resb);
  }
  return combine(resa,resb);
}
void updRUPQ(int l, int r, int val){
  for(l+=n,r+=n; l<r; l>>=1,r>>=1){
    if(l&1)t[l++] += val;
    if(r&1)t[--r] += val;
} }
int queryRUPQ(int pos, int res = 0){
  for(pos+=n; pos>0; pos>>=1) res += t[pos];
  return res;
}
void pushRUPQ(){ // inspect all elements: O(n)
  REP(i,1,n-1){
    t[i<<1] += t[i];  t[i<<1|1] += t[i];
    t[i] = 0;
} }
// h = height of the tree (log)
int h = sizeof(int) * 8 - __builtin_clz(n);
int d[N];
void apply(int p, int val){
  t[p] += val;  if(p<n) d[p] += val;
}
void build(int p){
  while(p>1){
    p>>=1;  t[p] = max(t[p<<1],t[p<<1|1]) + d[p];
} }
void push(int p){
  REV(s,h,1){
    int i = p>>s;
    if(d[i] != 0){
      apply(i<<1, d[i]);  apply(i<<1|1, d[i]);
      d[i] = 0;
} } }
void inc(int l, int r, int val){
  l+=n;  r+=n;  int lo=l, hi=r;
  for(; l<r; l>>=1, r>>=1){
    if(l&1) apply(l++, val);
    if(r&1) apply(--r, val);
  }
  build(lo);  build(hi-1);
}
int query(int l, int r){
  l+=n; r+=n;  push(l); push(r-1);
  int ret = -INF;
  for(; l<r; l>>=1, r>>=1){
    if(l&1) ret = max(ret, t[l++]);
    if(r&1) ret = max(ret, t[--r]);
  }
  return ret;
}
```

1.2 Persistent Segment Tree

```
class ST{ // example: persistent PURQ
  int n, posHead, posDat;
```

```
  int head[100005], dat[1800005];
  pii next[1800005]; // head * log(n)
  // private function
  int _upd(int,int,int,int,int);
  int _que(int,int,int,int,int);
public:
  int reset(int);
  int update(int);
  int query(int,int,int);
};
int _upd(int idx, int l, int r, int pos, int idx2){
  if(idx == idx2) return dat[idx];
  if(pos<l || r<pos) return dat[idx];
  if(l == r) return dat[idx] = 1;
  //
  pii& ptr = next[idx], ptr2 = next[idx2];
  ptr = ptr2;
  int mid = (l+r)>>1;
  if(pos <= mid)ptr.fi = ++posDat;
  else ptr.se = ++posDat;
  //
  int reta = _upd(ptr.fi,l,mid,pos,ptr2.fi);
  int retb = _upd(ptr.se,mid+1,r,pos,ptr2.se);
  return dat[idx] = reta + retb;
}
int ST::_que(int idx, int l, int r, int a, int b){
  if(r<a || b<l) return dat[0]; // default data
  if(a<=l&&r<=b) return dat[idx];
  //
  int mid = (l+r)>>1;  pii& ptr = next[idx];
  int reta = _que(ptr.fi,l,mid,a,b);
  int retb = _que(ptr.se,mid+1,r,a,b);
  return reta + retb;
}
int ST::reset(int _n){
  n = _n;  posDat = -1;
  head[posHead = 0] = ++posDat;
  dat[0] = 0;  next[0] = mp(0,0);
  return 0;
}
int ST::update(int pos){
  head[++posHead] = ++posDat;
  int prevHead = head[poshead-1];
  return _upd(head[posHead], 1,n, pos, prevHead);
}
int ST::query(int h, int l, int r){
  return _que(head[h], 1,n, l,r);
}
```

1.3 Binary Indexed (Fenwick) Tree

```
int n, dat[100005];
int LSOne(int n){ return (n&-n); }
void upd(int pos, int val){
  REPP(i,pos,n,LSOne(i)) dat[i] += val; // i|=i+1
}
int que(int pos, int res = 0){
  REVV(i,pos,1,LSOne(i)) res += dat[i];
  return res;  // i=(i&(i+1))-1
}
```

```
Range [l..r] Update
upd(BITree1, l, val);    upd(BITree2, l, val*(l-1));
upd(BITree1, r+1, -val); upd(BITree2, r+1, -val*r);
Range [1..k] Query
que(BITTree1, k)* k - que(BITTree2, k);
```

1.4 AVL Tree

```
Let:
z as first unbalanced node
y as larger height child of z
x as larger height child of y

Case:
z ->L y ->L x (LL Case): rotateRight(z)
z ->L y ->R x (LR Case): rotateLeft(y), rotateRight(z)
z ->R y ->L x (RL Case): rotateRight(y), rotateLeft(z)
z ->R y ->R x (RR Case): rotateLeft(z)
```

```
Rotation:
||       y                              x        ||
||      / \     Right Rotation         /  \      ||
||     x   T3   - - - - - - - - >    T1   y     ||
||    / \       < - - - - - - - -        / \    ||
||   T1  T2     Left Rotation          T2  T3    ||
```

1.5 Link Cut Tree

```
// LinkCut tree(n); // new tree with n nodes
// tree.link(x, y); // link x and y
// tree.cut(x);     // cut x
// tree.root(x);    // root of tree containing x
// tree.depth(x);   // depth of x in its tree
// tree.lca(x, y);  // LCA of x and y
struct Node{
  int sz, label; // size, label
  Node *p,*pp;   // parent, path-parent
  Node *l,*r;    // left, right pointers
  Node(){ p = pp = l = r =0; }
};
void update(Node *x){
  x->sz = 1;
  if(x->l)x->sz += x->l->sz;
  if(x->r)x->sz += x->r->sz;
}
void rotr(Node *x){
  Node *y,*z;  y = x->p, z = y->p;
  if((y->l = x->r))y->l->p = y;
  x->r = y, y->p = x;
  if((x->p = z)){
    if(y == z->l) z->l = x;
    else z->r = x;
  }
  x->pp = y->pp;  y->pp = 0;  update(y);
}
void rotl(Node *x){
  Node *y,*z;y = x->p, z = y->p;
  if((y->r = x->l))y->r->p = y;
  x->l = y, y->p = x;
  if((x->p = z)){
    if(y == z->l) z->l = x;
    else z->r = x;
  }
  x->pp = y->pp;  y->pp =0;  update(y);
}
void splay(Node *x){
  Node *y,*z;
  while(x->p){
    y = x->p;
    if(y->p ==0){
      if(x == y->l) rotr(x);
      else rotl(x);
    }
    else{
      z = y->p;
      if(y == z->l){
        if(x == y->l) rotr(y), rotr(x);
        else rotl(x), rotr(x);
      }
      else{
        if(x == y->r) rotl(y), rotl(x);
        else rotr(x), rotl(x);
  } } }
  update(x);
}
Node *access(Node *x){
  splay(x);
  if(x->r){
    x->r->pp = x; x->r->p = 0; x->r = 0; update(x);
  }
  Node *last = x;
  while(x->pp){
    Node *y = x->pp;  last = y;  splay(y);
    if(y->r) y->r->pp = y, y->r->p =0;
    y->r = x;  x->p = y;  x->pp =0;
    update(y);  splay(x);
  }
  return last;
}
```

```
Node *root(Node *x){
  access(x);  while(x->l) x = x->l;  splay(x);
  return x;
}
void cut(Node *x){
  access(x);  x->l->p = x->l = 0;  update(x);
}
void link(Node *x, Node *y){
  access(x);  access(y);  x->l = y;  y->p = x;
  update(x);
}
Node *lca(Node *x, Node *y){
  access(x);  return access(y);
}
int depth(Node *x){
  access(x);  return x->sz -1;
}
class LinkCut{
  Node *x;

  public:
  LinkCut(int n){
    x = new Node[n];
    FOR(i,n){ x[i].label = i;  update(&x[i]); }
  }
  virtual ~LinkCut(){ delete[] x; }
  void link(int u,int v){ ::link(&x[u],&x[v]); }
  void cut(int u){ ::cut(&x[u]); }
  int root(int u){ return ::root(&x[u])->label; }
  int depth(int u){ return ::depth(&x[u]); }
  int lca(int u, int v){
    return ::lca(&x[u],&x[v])->label;
} };
```

1.6 Tries

```
typedef struct{
  char kar;  bool isWord;  int parent;
  map<char,int> child;
} node;
char s[100];  node tree[2000001];
int idx,len,nn,ma;

void insert(int nIdx, int cIdx){ //nodeIdx,charIdx
  if(tree[nIdx].child.count(s[cIdx])==0){
    tree[++idx].kar = s[cIdx];
    tree[idx].parent = nIdx;
    tree[idx].child.clear();
    tree[nIdx].child[s[cIdx]] = idx;
  }
  if(cIdx == len-1)
    tree[tree[nIdx].child[s[cIdx]]].isWord=1;
  else insert(tree[nIdx].child[s[cIdx]],cIdx+1);
}
void traverse(int x, int depth){
  depth += tree[x].isWord;
  if(ma < depth)ma = depth, nn = x;
  for(auto i: tree[x].child)traverse(i.se,depth);
}
int main(){
  idx = 0;  tree[0].kar=0;  ma=-1;
  while(scanf("%s",s)){
    if(strcmp(s,".") == 0)break;
    len = strlen(s);
    insert(0,0);
  }
  nn = 0;
  traverse(0,0);
  int res[100], dd = 0;  char ss[100];
  while(nn){
    ss[dd] = tree[nn].kar;
    res[dd++] = tree[nn].isWord;
    nn = tree[nn].parent;
  }
  reverse(ss,ss+dd);  reverse(res,res+dd);
  ss[dd]=0;
  FOR(i,dd) if(res[i]){
    char temp = ss[i+1];  ss[i+1] = 0;
    printf("%s\n",ss);  ss[i+1] = temp;
} }
// Input   :   i if in input int integer output
// Output  :   i in int integer
```

```
// Another trie
#define MA 5000010
typedef struct Tnode{
  int counter;  char p;  int child[3];
} node;
char s[210];  node trie[MA];  int idx,len,res;

void insert(int trid, int id){
  trie[trid].counter++;
  if(id == len)return;
  if(trie[trid].child[s[id]-'0']==-1){
    trie[++idx].p = s[id];
    trie[trid].child[s[id]-'0'] = idx;
  }
  insert(trie[trid].child[s[id]-'0'],id+1);
}
void calc(int idx, int length){
  res = max(res,trie[idx].counter * length);
  if(trie[idx].child[0] != -1)
    calc(trie[idx].child[0],length+1);
  if(trie[idx].child[1] != -1)
    calc(trie[idx].child[1],length+1);
}
int main(){
  int t,n;
  scanf("%d",&t);
  while(t--){
    idx = 0;
    scanf("%d",&n);
    FOR(i,MA){
      trie[i].counter = 0;
      trie[i].child[0] = trie[i].child[1]=-1;
    }
    while(n--){
      scanf("%s", s);  len = strlen(s);
      insert(0,0);
    }
    res = 0;  calc(0,0);  printf("%d\n",res);
} }
```

1.7 Mo Algorithm

```
typedef pair<pii,int> iii;
int blocks,n,l,r,q,ans;
int answer[300100],cnt[1000100],a[300100];
vector<iii> v;

bool cmp(iii x, iii y){
  if(x.fi.fi/blocks == y.fi.fi/blocks)
    return x.fi.se<y.fi.se;
  return x.fi.fi/blocks < y.fi.fi/blocks;
}
void add(int position){
  if(++cnt[a[position]]==1)ans++;
}
void remove(int position){
  if(--cnt[a[position]]==0)ans--;
}
int main(){
  scanf("%d",&n);
  blocks =(int)sqrt(n);
  FOR(i,n)scanf("%d",&a[i]);
  scanf("%d",&q);
  FOR(i,q){
    scanf("%d%d",&l,&r); l--; r--;
    v.pb(mp(mp(l,r),i));
  }
  sort(v.begin(),v.end(),cmp);
  int currentL = 0, currentR = -1;
  FOR(i,(int)v.size()){
    l = v[i].fi.fi; r = v[i].fi.se;
    int idx = v[i].se;
    while(currentR < r)add(++currentR);
    while(currentR > r)remove(currentR--);
    while(currentL < l)remove(currentL++);
    while(currentL > l)add(--currentL);
    answer[idx]= ans;
  }
  FOR(i,q)printf("%d\n",answer[i]);
}
```

1.8 Misof Tree

```
class KthTree{
  private:
    int**tree, nlv;  int lv(int);
  public:
    KthTree(int);
    ~KthTree();
    void insert(int);
    void erase(int);
    int KthElement(int);
};
int KthTree::lv(int n, int ret = 0){
  while((1<<ret)<=n)ret++;  return ret;
}
KthTree::KthTree(int n){
  this->nlv = this->lv(n) + 1;
  tree=(int**)malloc(sizeof(int*)*(this->nlv));
  FOR(i,this->nlv){
    tree[i] = (int*)malloc(sizeof(int) * \
        (1<<(this->nlv-1)));
    memset(tree[i], 0, sizeof(int) * \
        (1<<(this->nlv-1)));
  }
}
KthTree::~KthTree(){
  FOR(i,this->nlv)free(tree[i]);  free(tree);
}
void KthTree::insert(int n){
  FOR(i,this->nlv)tree[i][n]++, n >>= 1;
}
void KthTree::erase(int n){
  FOR(i,this->nlv)tree[i][n]--, n >>= 1;
}
int KthTree::KthElement(int k){
  int a =0, b = (this->nlv)-1;
  while(b--) if(tree[b][a<<=1]<k)k -= tree[b][a++];
  return a;
}
```

1.9 Heavy-Light Decomposition

```
class Tree{
  int num, idx, n, elder;
  int dat[50005<<2];
  //LCA parent
  int par[50005][25];
  int depth[50005], hashh[50005];
  int chain[50005], headd[50005];
  vi child[50005];
  int count[50005]; // count subtree
  //generate function
  void genDepth(int,int);
  void genHLD(int);
  //LCA func
  int getNPar(int,int);
  int getAnc(int,int);
  //segment tree RURQ
  void upd(int,int,int,int,int,int);
  int que(int,int,int,int);
public:
  void reset(int);
  void push(int,int);
  void generate();
  void upd(int,int,int);
  int que(int);
};
void Tree::genDepth(int _node, int _depth=0){
  depth[_node] = _depth;  count[_node] = 1;
  for(auto x: child[_node]){
    if(x == par[_node][0])continue;
    par[x][0] = _node;
    genDepth(x,_depth+1);
    count[_node] += count[x];
} }
void Tree::genHLD(int _node){
  hashh[_node] = ++idx;
  chain[_node] = num;
  pii temp = mp(-1,-1);
  for(auto x: child[_node])
    if(count[x]>temp.se && x!=par[_node][0])
      temp=mp(x,count[x]);
  if(temp.fi != -1)genHLD(temp.fi);
  for(auto x: child[_node]){
    if(x==temp.fi || x==par[_node][0])continue;
    chain[x]=++num;  headd[num]=x;  genHLD(x);
} }
int Tree::getNPar(int _node, int _n){
  for(int j=0; j<15 && _n; j++, _n>>=1)
    if(_n&1)_node = par[_node][j];
  return _node;
}
int Tree::getAnc(int _u, int _v){
  //stabilize depth
  if(depth[_u]< depth[_v])
    _v = getNPar(_v,depth[_v]-depth[_u]);
  if(depth[_u]>depth[_v])
    _u = getNPar(_u,depth[_u]-depth[_v]);
  //u==v
  if(_u == _v)return _u;
  //search LCA
  FORD(j,20)
    if(depth[_u]>=(1<<j)&&par[_u][j]!=par[_v][j])
      _u = par[_u][j]; _v=par[_v][j];
  return par[_u][0];
}
void upd(int idx,int a,int b,int l,int r,int val){
  //base case
  if(b<l||r<a)return;
  if(l<=a&&b<=r){ dat[idx]+=val;  return;}
  //reccurence
  int mid = (a+b)>>1;
  upd(2*idx,a,mid,l,r,val);
  upd(2*idx+1,mid+1,b,l,r,val);
}
int Tree::que(int idx, int a, int b, int pos){
  //base case
  if(b<pos || pos<a)return 0;
  if(pos<=a && b<=pos)return dat[idx];
  //reccurence
  int mid = (a+b)>>1;
  if(pos <= mid)
    return dat[idx]+que(2*idx,a,mid,pos);
  return dat[idx]+que(2*idx+1,mid+1,b,pos);
}
//public
void Tree::reset(int _n){
  n = _n;
  FOR(i,_n+3){
    FOR(j,23)par[i][j] = 0;
    child[i].clear();
    hashh[i] = depth[i] = count[i] = 0;
    chain[i] = headd[i] = -1;
  }
  FOR(i,_n<<2)dat[i] = 0;
  num = idx = elder = 0;
}
void Tree::push(int _u, int _v){
  child[_u].pb(_v);  child[_v].pb(_u);
}
void Tree::generate(){
  //generate LCA
  genDepth(elder);
  REP(j,1,20) REP(i,0,n-1)
    par[i][j] = par[ par[i][j-1]][j-1];
  //generate HLD :)
  chain[elder] = ++num;  headd[num] = elder;
  genHLD(elder);
}
void Tree::upd(int _u, int _v, int _val){
  int anc = getAnc(_u,_v);
  while(1){ // run from U
    int fron = headd[chain[_u]];
    if(chain[anc] == chain[_u])fron=anc;
    upd(1,1,n,hashh[fron],hashh[_u],_val);
    if(chain[anc] == chain[_u])break;
    _u = par[fron][0];
  }
  while(1){ // run from V
    int fron = headd[chain[_v]];
    if(chain[anc] == chain[_v])fron=anc;
    upd(1,1,n,hashh[fron],hashh[_v],_val);
    if(chain[anc] == chain[_v])break;
    _v = par[fron][0];
  }
```

```
  //substract anc
  upd(1,1,n,hashh[anc],hashh[anc],-_val);
}
int Tree::que(int pos){
  return que(1,1,n,hashh[pos]);
}
```

2 Dynamic Programming

2.1 LIS (Longest Increasing Subsequence)

```
vi indi, arr((int)1e6,0);  arr[0]=INF;  int cnt=0;

FOR(i,n){
  int lo = 0, hi = cnt, mid;
  while(lo<=hi){
    mid = (lo+hi)/2;
    if(arr[mid] >= lst[i])hi = mid-1;
    else lo = mid+1;
  }
  arr[lo] = lst[i];  indi.pb(lo);
    if(cnt<lo)cnt=lo;
}
vi ans;  int idx = n-1;
while(cnt>0){
  if(indi[idx]==cnt)ans.pb(lst[idx]), cnt--;
  idx--;
}
reverse(ans.begin(),ans.end());
```

2.2 Cyclic LCS (Longest Common Subsequence)

```
// O(nm)   https://arxiv.org/pdf/1208.0396v3.pdf
const int N = 4000+10;
int dp[N][N], from[N][N];
int eq(int a, int b){
  return s[(a-1)%n]==t[(b-1)%m];
}
int clcs(){
  REP(i,0,n<<1) REP(j,0,m)
    dp[i][j] = from[i][j] = 0;
  REP(i,0,n<<1) REP(j,0,m){
    dp[i][j] = 0;
    if(j && dp[i][j-1] > dp[i][j])
      dp[i][j] = dp[i][j -1], from[i][j]=0;
    if(i&&j && eq(i,j) && dp[i-1][j-1]+1>dp[i][j])
      dp[i][j] = dp[i-1][j-1]+1, from[i][j]=1;
    if(i && dp[i-1][j] > dp[i][j])
      dp[i][j] = dp[i-1][j], from[i][j]=2;
  }
  int ret = 0;
  FOR(i,n){
    ret = max(ret,dp[i+n][m]);
    int x = i+1, y = 0;
    while(y<=m && from[x][y]==0) y++;
    for(; y<=m && x<=n*2; x++){
      from[x][y] = 0;  dp[x][m]--;
      if(x == n*2)break;
      for(; y <= m; y++){
        if(from[x+1][y] == 2)break;
        if(y+1<= m && from[x+1][y+1]==1){
          y++;  break;
  } } } }
  return ret<<1;
}
```

2.3 Longest Common Increasing Subsequence

```
const int OO =(int)1e9;  const ll INF =(ll)1e18;
int getint(){ int x;  scanf("%d",&x);  return x; }
const int MAXN =512;
int A[MAXN];  int na;  int B[MAXN];  int nb;
int prev[MAXN][MAXN];
ll best[MAXN][MAXN];
int pos[MAXN][MAXN];

void printPath(int path){
  if(prev[path>>10][path&1023] == -1)
    printf("%d",A[path>>10]);
  else{
    printPath(prev[path>>10][path&1023]);
    printf(" %d",A[path>>10]);
} }
int main(){
  FOR(i,na=getint())A[i] = getint();
  FOR(i,nb=getint())B[i] = getint();
  FOR(i,nb) FOR(j,nb)best[i][j] = INF;
  FOR(i,na) FOR(j,nb)prev[i][j] = -1;
  FOR(i,na){
    int x = -1, p =0;
    FOR(j,nb){
      if(A[i] == B[j]){
        int y = p;  while(best[j][y]<A[i])y++;
        best[j][y] = A[i];  pos[j][y] = (i<<10)+j;
        if(y >0)prev[i][j] = pos[j][y-1];
        x = p = y;
      } else if(x!=-1 && best[j-1][x]<best[j][x]){
        best[j][x] = best[j-1][x];
        pos[j][x] = pos[j-1][x];
      } else x = -1;
  }   }
  int len = -1;
  FORD(i,nb) if(best[nb-1][i]!=INF){
    len=i;  break;
  }
  printf("%d\n",len+1);
  if(len >=0)printPath(pos[nb-1][len]);  puts("");
}
```

2.4 DP Buka String dan Lipat String

```
struct tipetipe{ int min, nkurung; };  string s2;
tipetipe memo[305][305];  int n, path[305][305][4];

string buka(int& i, string s){
  int len = s.length(), angka=0;  string tmp;
  while(i<len && s[i]!=')'){
    if(s[i]>='0'&& s[i]<='9')
      angka = angka*10+(s[i]-'0');
    else if(s[i]=='('){
      string tmp2 = buka(++i,s);
      REP(i,angka)tmp += tmp2;
      angka = 0;
    } else tmp = tmp+s[i];
    i++;
  }
  return tmp;
}
tipetipe hohoho(int l, int r){
  // printf("%d %d\n",l,r);  LOL;  STOP;
  if(memo[r][l].min != -1)return memo[r][l];
  tipetipe ret;  ret.min = r-l+1;  ret.nkurung = 0;
  path[l][r][0] = 0;
  REP(i,1,(r-l+1)/2) if((r-l+1)%i == 0){
    string t1 = s2.substr(0+l,i);
    string t2 = s2.substr(i+l,i);
    int ni = 1, j = i+l;
    while(t1==t2 && j<=r){
      ni++;  j+=i;  t2 = s2.substr(j,i);
    }
    if(ni == (r-l+1)/i){
      int dig = 0, nii = ni;
      while(ni>0)dig++, ni /= 10;
      tipetipe tmp = hohoho(l,l+i-1);
      tmp.min += dig+2;  tmp.nkurung += 2;
      if(tmp.min < ret.min){
        ret = tmp;
        path[l][r][0] = 1;
        path[l][r][1] = nii;
        path[l][r][2] = i;
      } else if(tmp.min==ret.min && \
          tmp.nkurung < ret.nkurung){
        ret = tmp;  path[l][r][0]=1;
        path[l][r][1]= nii;     path[l][r][2]= i;
  } } }
  REP(i,l,r-1){
    tipetipe tmp, left, right;
    left = hohoho(l,i);  right = hohoho(i+1,r);
    tmp.min = left.min + right.min;
    tmp.nkurung = left.nkurung + right.nkurung;
    if(tmp.min< ret.min){
      ret = tmp; path[l][r][0]=2; path[l][r][1]=i;
```

```
    } else if(tmp.min== ret.min && \
        tmp.nkurung < ret.nkurung){
      ret = tmp; path[l][r][0]=2; path[l][r][1]=i;
    }
  }
  // printf("l->%d r->%d res->%d path->%d\n", \
  l,r,ret,path[l][r][0] );
  return memo[r][l]= ret;
}
string search_path(int l, int r){
  string ret;
  if(path[l][r][0]==0) FOR(i,l,r)ret += s2[i];
  if(path[l][r][0]==1){
    char tmpx[10];
    sprintf(tmpx,"%d",path[l][r][1]);  ret += tmpx;
    ret+="("+search_path(l,l+path[l][r][2]-1)+")";
  }
  if(path[l][r][0]==2){
    ret += search_path(l,path[l][r][1]);
    ret += search_path(path[l][r][1]+1,r);
  }
  return ret;
}
int main(){
  cin >> n;
  FOR(i,n){
    string s1;  cin >> s1;
    int z = 0;  s2 = buka(z,s1);
    memset(memo,-1,sizeof(memo));
    hohoho(0,s2.length()-1);
    cout << s2 << endl;
    cout << search_path(0,s2.length()-1) << endl;
} }
// input
// 2(haue)yueuyeuue3(xxzz)zzxx
// hauehaueyueuyeuuexxzzxxzzxxzzzzxx
// 2(haue)yueuyeuue3(xxzz)zzxx
```

2.5 Matrix Chain Multiplication

```
const int MAX=100;  const int INF=(int)2e9;
int p[MAX+1], m[MAX+1][MAX+1], s[MAX+1][MAX+1];

void MCM(int n){
  FOR(i,n+1)m[i][i]=0;
  REP(l,2,n)REP(i,1,n-l+1){
    Int j = i+l-1;   m[i][j] = INF;
    REP(k,i,j-1){
      int q = m[i][k]+m[k+1][j]+p[i-1]*p[k]*p[j];
      if(q < m[i][j])m[i][j] = q, s[i][j] = k;
} } }
void print_MCM(int i, int j){
  if(i == j)printf("A%d",i);
  else{
    printf("(");  print_MCM(i,s[i][j]);
    printf(" x ");  print_MCM(s[i][j]+1,j);
    printf(")");
} }
int main(){
  int n, i, j, cs = 1;
  while(scanf("%d",&n) == 1){
    if(n == 0)break;
    FOR(i,n)scanf("%d%d",&p[i],&p[i+1]);
    printf("Case %d: ",cs++);  MCM(n);
    print_MCM(1,n);  puts("");
} }
```

2.6 Min Cut Polygon Triagulation

```
int getCost(pii a, pii b, pii c){
  return getDist2(a,b)+getDist2(b,c)+getDist2(c,a);
}
int minCutPolygonTriagulation(){
  int res=0, dp[105][105];
  FOR(i,103) FOR(j,103)dp[i][j] = 0;
  FOR(i,n)res += getDist2(dat[i],dat[(i+1)%n]);
  //DP
  FOR(xx,n) for(int i=0,j=xx; j<n; i++,j++){
    dp[i][j] = (j<i+2) ? 0 : INF;
    REP(k,i+1,j-1) dp[i][j] = min( \
      dp[i][j], dp[i][k]+dp[k][j] + \
      getCost(dat[i],dat[j],dat[k]) );
  }
  return(dp[0][n-1]-res)>>1;
}
```

2.7 Kadane

```
// Kadane 1D
k = 0;  l = 0;  s = -INF;  t = 0;  j = 1;
REP(i,1,n){
  t += a[i];
  if(t > s)k = j, l = i, s = t;
  if(t < 0)t = 0, j = i+1;
}
// Kadane 2D
int N, t =0, a[100][100], pr[100];
int S = 1<<31, s=0, k, l, x1=0,x2=0, y1=0,y2=0, j;

FOR(z,N){
  FOR(i,N)pr[i] = 0;
  REP(i,z,N-1){
    t = j = k = l = 0, s =1<<31;
    FOR(i,n){
      pr[i] = pr[i]+ a[x][i];
      t = t + pr[i];
      if(t > s)s = t, k = i, l = j;
      if(t < 0)t = 0, j = i+1;
    }
    if(s > S)S = s, x1 = x, y1 = k, x2 = z, y2 = l;
} }
cout << x1 <<" "<< y1 <<" "<< x2 <<" "<< y2 <<endl;
cout << S;
```

2.8 DP SOS

```
// Given array A[] of 2^N integers, find sum of all
// F(x), where F(x) = sum of all A[i] where x&i=i

// O(3^N)  This one can be used for various other
// problems. Flexible;  iterate over all the masks
FOR(mask,(1<<n)){
  F[mask] = A[0];
  // iterate over all the subsets of the mask
  for(int i=mask; i; i=(i-1)&mask) F[mask] += A[i];
}
// O(N * 2^N)  memory optimized, easy to code.
FOR(i,(1<<N))F[i] = A[i];
FOR(i,N) FOR(mask,(1<<N)) if(mask&(1<<i))
  F[mask] += F[mask^(1<<i)];
```

2.9 DP Convex Hull

```
// Name       : Convex Hull Optimization 1
// Recurrence : dp[i] = min(dp[j] + b[j]*a[i])
// Condition  : b[j] >= b[j+1] and a[i] <= a[i+1]
// Complexity : O(n^2) -> O(n)

// Name       : Convex Hull Optimization 2
// Recurrence : dp[i][j]=min(dp[i-1][k]+b[k]*a[j])
// Condition  : b[k] >= b[k+1] and a[j] <= a[j+1]
// Complexity : O(kn^2) -> O(kn)
struct line{
  ll a,b;  ll getval(ll x){ return a*x+b; }
};
ll xsmall(line l1, line l2, line l3){
  ll x1 = (l3.b-l1.b)*(l1.a-l2.a);
  ll x2 = (l2.b-l1.b)*(l1.a-l3.a);
  return x1<x2;
}
void getpoint(int k){
  v.clear();
  REP(i,1,n){
    line temp;  temp.a = -i;
    temp.b = (i*s[i]) - sum[i] + dp[i][k-1];
    while(v.size()>=2 && \
      xsmall(v[v.size()-2],v.back(),temp)) v.pob();
    v.pb(temp);
} }
int main(){
  scanf("%d%d",&n,&k);
  REP(i,1,n){
```

```
    scanf("%lld",&f[i]);
    sum[i] = sum[i-1]+(f[i]*(i-1));
    s[i] = s[i-1]+f[i];
  }
  REP(i,1,n)dp[i][1] = sum[i];  int pt = 0;
  REP(i,2,k){
    getpoint(i);  pt = 0;
    REP(j,1,n) while(pt<v.size()-1 && \
        v[pt].getval(s[j]) > v[pt+1].getval(s[j]))
      pt++;
    dp[j][i] = sum[j]+v[pt].getval(s[j]);
  }
  printf("%lld\n",dp[n][k]);
}
```

2.10 DP Divide and Conquer

```
// dp[i][j] = min(dp[i-1][k] + C[k][j]) for (k<j)
// A[i][j] <= A[i][j+1]
// O(knlogn)

void solve(int k, int l, int r, int L , int R){
  if(l > r)return;
  int mid = (l+r)/2, opt = -1;
  REP(i,L,min(R,mid)){
    ll tmp = dp[k-1][i]+cost[i+1][mid];
    if(tmp < dp[k][mid])dp[k][mid] = tmp, opt = i;
  }
  solve(k,l,mid-1,L,opt);  solve(k,mid+1,r,opt,R);
}
int main(){
  // init array of dp to infinity
  // init array dp[1][i] = to cost[1][i]
  REP(i,2,k)solve(i,1,n,1,n);
}
```

2.11 DP Knuth Yao

```
// SPOJ BREAKING STRINGS
// dp[i][j] = min(dp[i][k]+dp[k][j]) + C[i][j]
// for (i<k<j) and A[i,j-1] <= A[i,j] <= A[i+1,j]
// O(n^2)
ll n, dp[1005][1005], f[1005];
int m, mid[1005][1005];

int main(){
  while(scanf("%lld%d",&n,&m) == 2){
    REP(i,1,m)scanf("%lld",&f[i]);
    f[0] = 0;  f[m+1] = n;
    FOR(i,m+2) FOR(j,m+2)
      dp[i][j] = 1e18, mid[i][j] = i;
    FOR(s,m+2) FOR(l,m-s+2){
      int r = l+s;
      if(s < 2){
        dp[l][r] = 0;  mid[l][r] = l;  continue;
      }
      int mleft = mid[l][r-1];
      int mright = mid[l+1][r];
      dp[l][r] = 1e18;
      REP(i,mleft,mright){
        ll tmp = dp[l][i] + dp[i][r] + f[r]-f[l];
        if(tmp < dp[l][r])
          dp[l][r] = tmp, mid[l][r] = i;
    } }
    printf("%lld\n",dp[0][m+1]);
} }
```

3 Graphs

3.1 Link Cut Tree

```
void topoSort(int v){
  vis[v] = 1;
  FOR(i,(int)ad[v].size())
    if(!vis[ad[v][i]]) topoSort(ad[v][i]);
  topoSorted.pb(v);
}
int main(){
  topoSorted.clear();
  REP(i,N) if(!vis[i]) topoSort(i);
  reverse(topoSorted.begin(),topoSorted.end());
}
```

Kahn's Algorithm

```
vi topoSortKahn(){
  // get all in-degrees
  vi in_degree(V,0);
  FOR(u,V) for(int x: adj[u]) in_degree[x]++;
  queue<int> q;
  // queue all indegree == 0
  FOR(i,v) if(in_degree[i] == 0) q.push(i);
  int cnt = 0;  vi res;
  while(!q.empty()){
    int u = q.front();  q.pop();  res.pb(u);
    for(int x: adj[u])
      if(--in_degree[x] == 0)q.push(x);
    cnt++;
  }
  // Check if there was a cycle
  if(cnt != V){ res.clear(); puts("Cycle!"); }
  return res;
}
```

3.2 k-th Shortest Path

```
// k = shortest path ke berapa.
// n = jumlah vertex.
// solve(i,j); shortest path dari i ke j.
const int MAXN = 100;  const int MAXK = 40;
vii g[MAXN];  priority_queue<pii>pq;
int visit[MAXN], n, k;

int solve(int s, int t){
  FILL(visit,0);  pq.push(MP(0,s));  int ret = 0;
  while(!pq.empty()){
    int awal = -pq.top().fi;
    int f = pq.top().se; pq.pop();
    if(visit[f]>=k)continue;
    visit[f]++;
    if(f==t)ret+=awal;
    for(auto x: g[f]) if(visit[x.fi]<k)
      pq.push(MP(-(awal+x.se),x.fi));
  }
  return ret;
}
int main(){
  k = GI;  n = GI;
  FOR(nedge,GI){
    int a = GI-1, b = GI-1, c = GI;
    g[a].pb(mp(b,c));  g[b].pb(mp(a,c));
  }
  printf("%d\n",solve(0,n-1));
}
```

3.3 Manhattan Spanning Tree

```
const int INF =(int)1e9;
struct Pos{ int x,y, diff,sum, id; } p[150000];
vi BIT;  vector<pair<pair<int,int>,int>> Raw_Edges;

bool sortpoint(Pos A, Pos B){
  if(A.x != B.x)return A.x > B.x;
  return A.y > B.y;
}
int GetNearest(int x){
  int ret =-1;
  for(; x>0; x-=(x&(-x)))
    if(BIT[x]!=-1&&(ret==-1 || \
      p[BIT[x]].sum < p[ret].sum)) ret = BIT[x];
  return ret;
}
void Insert(int x,int id){
  int maxi = BIT.size();
  for(; x<maxi; x+=(x&(-x)))
    if(BIT[x]==-1 || (p[id].sum<p[BIT[x]].sum))
      BIT[x] = id;
}
void AddEdges(){
  viaxis;  axis.pb(-INF);
  FOR(a,n)axis.pb(p[a].diff);
  sort(axis.begin(),axis.end());
  axis.erase( unique(axis.begin(),axis.end()), \
```

```
    axis.end() );
  sort(p, p+n, sortpoint);
  BIT = vi(axis.size(),-1);
  FOR(a,n){
    int pos = lower_bound(axis.begin(), \
      axis.end(), p[a].diff) - axis.begin();
    int nearest = GetNearest(pos);
    if(nearest != -1) Raw_Edges.pb(mp(mp(p[a].id, \
      p[nearest].id),p[nearest].sum - p[a].sum));
    Insert(pos,a);
} }
int main(){
  FOR(a,n){
    scanf("%d%d",&p[a].x,&p[a].y);
    p[a].sum = p[a].x + p[a].y;
    p[a].diff = p[a].x - p[a].y;  p[a].id = a;
  }
  FOR(a,4){
    FOR(b,n){
      if(a==1 || a==3){
        swap(p[b].x,p[b].y);
        p[b].diff = p[b].x - p[b].y;
      } else if(a==2){
        p[b].x =-p[b].x;
        p[b].sum = p[b].x + p[b].y;
        p[b].diff = p[b].x - p[b].y;
      }
    } AddEdges();
  }
  //DO MST
}
```

3.4 Directed MST

```
int con[MAXN][MAXN], in[MAXN];
int vis[MAXN], active[MAXN], n;
//state: 0:unvisited, 1:visited, 2:cycle, 3:active
int cycle(int idx, int& res){
  if(vis[idx] == 0){
    vis[idx] = 3;
    if(in[idx] == -1) vis[idx] = 1;
    else vis[idx] = cycle(in[idx],res);
  } else if(vis[idx] == 3){
    set<int> pseudoNodes;  int minCycle = INF;
    for(int j=idx; ; j=in[j]){
      pseudoNodes.insert(j);
      minCycle <?= con[in[j]][j];
      if(in[j]==idx)break;
    }
    int minCost = INF, bestParent;
    for(auto j: pseudoNodes){
      assert(active[j]);
      FOR(i,n) if(active[i] && con[i][j]!=-1 && \
      !pseudoNodes.count(i)){
        if(minCost > con[i][j]-con[in[j]][j]){
          minCost = con[i][j]-con[in[j]][j];
          bestParent = i;
        }
        con[i][j] -= con[in[j]][j]-minCycle;
    } }
    res += minCost;
    in[idx] = bestParent;
    for(auto j: pseudoNodes){
      FOR(i,n) if(active[i] && i!=idx && \
        con[i][j]!=-1) if(con[i][idx]==-1 || \
        con[i][idx]>con[i][j])
          con[i][idx] = con[i][j];
      FOR(i,n)
        if(active[i]&&i!=idx&&con[j][i]!=-1){
          if(in[i]==j)in[i] = idx;
          if(con[idx][i]==-1 || \
              con[idx][i] > con[j][i])
            con[idx][i]= con[j][i];
        }
        if(j != idx)active[j] = false;
    }
    vis[idx] = 2;
  }
  return vis[idx];
}
int dmst(int root){
  int res = 0;  in[root] = -1;
  FOR(i,n)con[i][root] = -1;
  FOR(i,n) if(i!=root){
    int minCost = INF;
    FOR(j,n) if(con[j][i]!=-1 && con[j][i]<minCost)
      minCost = con[j][i], in[i] = j;
    res += minCost;
  }
  memset(active,1,sizeof(active));
  while(1){
    memset(vis,0,sizeof(vis));
    bool hasCycle = cycle(root,res)==2;
    FOR(i,n) if(active[i] && !vis[i])
      hasCycle |=(cycle(i,res)==2);
    if(!hasCycle)return res;
} }
int dfs(int idx){
  if(vis[idx])return 0;
  int ret = 1;  vis[idx] = true;
  FOR(i,n) if(con[idx][i] != -1)ret += dfs(i);
  return ret;
}
bool connected(int root){
  memset(vis,0,sizeof(vis));  return dfs(root)==n;
}
int main(){
  int nTC, m, u, v, w;
  scanf("%d",&nTC);
  REP(TC,1,nTC){
    printf("Case #%d: ",TC);
    scanf("%d%d",&n,&m);
    memset(con,-1,sizeof(con));
    FOR(i,m)scanf("%d%d%d",&u,&v,&w), con[u][v]= w;
    if(!connected(0))puts("Possums!");
    else printf("%d\n",dmst(0));
} }
```

3.5 Bellman Ford (Negative Cycle Detection)

```
int bellmanFord(int s){
  int i, j;  FOR(i,n)d[i]= INF;  d[s]=0;
  FOR(i,n-1) FOR(j,e)
    if(d[edge[j].v] > d[edge[j].u]+edge[j].w)
      d[edge[j].v] = d[edge[j].u]+edge[j].w;
  FOR(j,e)
    if(d[edges[j].u] + edges[j].w < d[edges[j].v])
      return 0;
  return 1;
}
```

3.6 Dinic Algorithm

```
struct Edge{
  int v, flow, c, rev;
  Edge(int _v, int _flow, int _c, int _rev){
    v=_v;  flow=_flow;  c=_c;  rev=_rev;
  }
}
class Graph{
private:
  int V, lvl[5005], start[5005];
  vector<Edge> adj[5005];
public:
  Graph(int V){ this->V = V; }
  void addEdge(int u, int v, int C){
    Edge a(v,0,C,adj[v].size());
    Edge b(u,0,0,adj[u].size());
    adj[u].pb(a);
    adj[v].pb(b);
  }
  bool BFS(int s, int t){
    memset(lvl,-1,sizeof(lvl));
    lvl[s] = 0;
    queue<int> q; q.push(s);
    while(!q.empty()){
      int u=q.front();  q.pop();
      for(auto e: adj[u]){
        if(lvl[e.v]<0 && e.flow<e.c){
          lvl[e.v] = lvl[u]+1;
          q.push(e.v);
    } } }
    return lvl[t]!=-1;
```

```
  }
  int DFS(int u,int flow,int t,int start[]){
    if(u==t)return flow;
    for(; start[u]<adj[u].size(); start[u]++){
      Edge &e = adj[u][start[u]];
      if(lvl[e.v]==lvl[u]+1 && e.flow<e.c){
        int curr = min(flow,e.c-e.flow);
        int tmp = DFS(e.v,curr,t,start);
        if(tmp>0){
          e.flow+=tmp;
          adj[e.v][e.rev].flow-=tmp;
          return tmp;
    } } }
    return 0;
  }
  int dinic(int s, int t){
    int total=0;
    while(BFS(s,t)){
      memset(start,0,sizeof(start));
      while(int flow=DFS(s,INT_MAX,t,start))
        total+=flow;
    } return total;
} }
```

3.7 LowestCommonAncestor (SegTree)

```
const int MAXN = 200005;  const int INF = (int)1e9;
int n, q, euler[2*MAXN];
int level[2*MAXN], h[2*MAXN], id;
// h[i] = idx of the first occurence of node i in E
vi adj[maxn]; bool vis[maxn];
struct node{int level,id;} st[4*2*MAXN];

void eulerTour(int u, int p, int l){
  euler[id] = u;  level[id] = l;
  if(!vis[u])vis[u] = 1, h[u] = id;
  id++;
  for(auto v: adj[u]) if(v != p){
    eulerTour(v, u, l+1);
    euler[id] = u;  level[id++] = l;
} }
void build(int p, int l, int r){
  if(l == r){
    st[p].level = level[l]; st[p].id = l; return;
  }
  int mid = (l+r)/2;
  build(p*2, l, mid);  build(p*2+1, mid+1, r);
  st[p].level = min(st[p*2].level,st[p*2+1].level);
  st[p].id = (st[p*2].level < st[p*2+1].level) ? \
    st[p*2].id : st[p*2+1].id;
}
node query(int p,int l,int r, inti,int j){
  if(l > j || r <i){
    node ret;  ret.level = INF;  return ret;
  }
  if(l >= I && r <= j)return st[p];
  int mid = (l+r)/2;
  node ret1 = query(p*2,l,mid,i,j);
  node ret2 = query(p*2+1,mid+1,r,i,j);
  return (ret1.level < ret2.level) ? ret1 : ret2;
}
int lca(int r,int u,int v){
  int a = min(h[u],h[v]), b = max(h[u],h[v]);
  int lca_id = query(1,1,2*n-1,a,b).id;
  return euler[lca_id];
}
int main(){
  int r, u, v;
  scanf("%d",&n);
  FOR(i,n-1){
    scanf("%d%d",&u,&v);
    adj[u].pb(v); adj[v].pb(u);
  }
  memset(euler,0,sizeof(euler));
  memset(level,0,sizeof(level));
  memset(h,0,sizeof(h));
  memset(vis,0,sizeof(vis));
  id =1;  eulertour(1,1,0);
  memset(st,0,sizeof(st));
  build(1,1,2*n-1);
  scanf("%d",&q);
  while(q--){
    scanf("%d%d%d",&r,&u,&v);
    int lca1, lca2, lca3, ans;
    lca1 = lca(1,r,u);
    lca2 = lca(1,r,v);
    lca3 = lca(1,u,v);
    if(lca1 == lca2 && lca2 == lca3) ans = lca1;
    else if(lca1==lca2 && lca3!=lca1) ans = lca3;
    else if(lca1==lca3 && lca2!=lca1) ans = lca2;
    else if(lca2==lca3 && lca1!=lca2) ans = lca1;
    printf("%d\n",ans);
} }
```

Let LCA(r,u,v) be the LCA of u and v with respect to root r. To compute LCA(r,u,v), we can compute, for any fixed ro (usually let ro = 1),
LCA(ro,r,u); // LCA(1,r,u);
LCA(ro,r,v); // LCA(1,r,v);
LCA(ro,u,v); // LCA(1,u,v);
and take the "odd man out", i.e., if two are equal and the third are different, then take the third, else if they are all equal, take any one.

3.8 Min Cost Max Flow

```
bool bellman(){
  FOR(i,260) dist[i]=INF, p[i]=-1;
  dist[src] = 0; bool alter;
  FOR(k,259){
    alter = false;
    FOR(i,260){
      for(auto v: adj[i]){
        int w = cost[i][v];
        if(dist[v]>dist[i]+w && res[i][v]){
          alter = true;
          dist[v] = dist[i]+w;
          p[v] = i;
    } } }
    if(!alter) break;
  } return (dist[sink] != INF);
}
void augment(int u,int minEdge){
  if(src == u){ flow = minEdge; return; }
  else if(p[u]!=-1){
    augment(p[u],min(minEdge,res[p[u]][u]));
    res[p[u]][u] -= flow;
    res[u][p[u]] += flow;
    costPerFlow += cost[p[u]][u];
} }

// inside int main
// construct graph
// make bidirectional edge from src to i
cost[src][i] = x; cost[i][src] = -x;
res[src][i] = y;
// make bidirectional edge from i to sink
cost[i][sink] = x;  cost[sink[i]] = -x;
res[i][sink] = y;
// do it
while(bellman()){
  flow=0;
  augment(sink,INF);
  if(flow==0) breakl;
  mf += flow;
  totCost += costPerFlow;
}
```

3.9 Min Cost Max Flow (Multiple Edges)

```
typedef struct{ int v, flow, cost, cap; } node;
node parent[MAX_N+1];  vector<node> adj[MAX_N+1];
int dist[MAX_N+1], from[MAX_N+1], v, e;

int augment(int s,int t){
  bool update;
  REP(i,1,v)dist[i] = INF, from[i] = -1;
  dist[s]=0;
  REP(i,1,v-1){ //bellman ford
    update = false;
    REP(j,1,v) FOR(k,adj[j].size())
    if(adj[j][k].flow < adj[j][k].cap){
```

```
      int edgeCost = adj[j][k].cost;
      if(adj[j][k].flow <0) edgeCost = -edgeCost;
      if(dist[adj[j][k].v] > dist[j]+edgeCost){
        dist[adj[j][k].v] = dist[j]+ edgeCost;
        parent[adj[j][k].v] = adj[j][k];
        from[adj[j][k].v] = j;
        update = true;
    } } if(!update)break;
  }
  if(dist[t]==INF)return 0;
  int p = t, f = INF, pa;  node temp;
  for(; from[p]!=-1; p=from[p]){
    pa = from[p];  temp = parent[p];
    f = min(f,temp.cap-temp.flow);
  }
  p = t;
  for(; from[p]!=-1; p=from[p]){
    pa = from[p];
    FOR(i,adj[pa].size()){
      if(adj[pa][i].v == p && \
          adj[pa][i].cost == parent[p].cost){
        adj[pa][i].flow += f; break;
    } }
    FOR(i,adj[p].size()){
      if(adj[p][i].v == pa && \
          adj[p][i].cost == parent[p].cost){
        adj[p][i].flow -= f; break;
  } } }
  return f*dist[t];
}
void addedge(int v1, int v2, int c, int cap, \
    int flow=0){
  bool f = true;
  FOR(i,adj[v1].size())
    if(adj[v1][i].v==v2 && adj[v1][i].cost==c){
      adj[v1][i].cap += cap;
      adj[v1][i].flow += flow;
      f = false;  break;
    }
  if(f) adj[v1].pb((node){v2,flow,c,cap});
}
//Add undirected edge a-b dengan cost c, cap 1
addedge(a,b,c,1);  addedge(b,a,c,1);
//directed edge IN(i)->OUT(i) dengan cost a, cap 1
addedge(IN(i),OUT(i),-a,1,0);
addedge(OUT(i),IN(i),a,1,1); //blocking flow
```

3.10 Max Flow (Adj List)

```
const int maxv = 100, INF = (int)1e9;
int res[maxv][maxv], s, t, mf, f;  vi p, adj[maxv];

void augment(int v, int minedge){
  if(v == s){ f = minedge; return; }
  else if(p[v]!=-1){
    augment(p[v], min(minedge, res[p[v]][v]));
    res[p[v]][v]-= f;
    res[v][p[v]]+= f;
} }
int main(){
  mf = 0;
  while(1){ // BFS
    f = 0;  vi dist(maxv,INF); dist[s]=0;
    queue<int> q;  q.push(s);  p.assign(maxv,-1);
    while(!q.empty()){
      int u = q.front();  q.pop();
      if(u == t)break;
      for(auto v: adj[u])
        if(dist[v]==INF && res[u][v]>0)
          dist[v] = dist[u]+1, p[v] = u, q.push(v);
    }
    augment(t,INF);  mf += f;  if(f == 0)break;
  }
  printf("%d\n",mf);
}
```

3.11 Min Cut (Stoer Wagner)

```
int m[600][600], node[600], inset[600], d[600];

int reduce(int n){
  memset(inset,0,sizeof(inset));
  memset(d,0,sizeof(d));
  inset[node[0]] = 1;
  REP(i,1,n-1)d[node[i]] = m[node[0]][node[i]];
  int ltnode = 0;
  REP(_,1,n-1){
    int nd = -1;
    REP(i,1,n-1){
      if(inset[node[i]])continue;
      if(nd ==-1)nd = i;
      if(d[node[i]]> d[node[nd]])nd = i;
    }
    inset[node[nd]] = 1;
    if(_ == n-1){
      FOR(i,n){
        Int xxa = node[ltnode], xxb= node[i];
        m[xxa][xxb] += m[node[nd]][node[i]];
        m[xxb][xxa] = m[xxa][xxb];
      }
      int ret = d[node[nd]];  node[nd] = node[n-1];
      return ret;
    }
    ltnode = nd;
    FOR(i,n){
      if(inset[node[i]])continue;
      d[node[i]]+= m[node[nd]][node[i]];
} } }
int mincut(int n){
  FOR(i,n) node[i]= i;
  int ans = (1<<30);
  while(n > 1){
    int mcut = reduce(n);
    if(mcut < ans) ans = mcut;
    n--;
  }
  return ans;
}
int main(){
  int N, M;
  while(scanf("%d%d",&N,&M)!=EOF){
    memset(m, 0, sizeof(m));
    FOR(i,M){
      int a,b,c; scanf("%d%d%d",&a,&b,&c);
      m[a][b] += c;  m[b][a] += c;
    }
    printf("%d\n", mincut(N));
} }
```

3.12 Eulerian Tour Checking

```
// Every edge is visited exactly once
int N, M, d[250], di[250];
int m[205][205], pre[205],vis[205];

int flow(int f, int si){
  queue<int> q;  q.push(f);
  memset(vis,0,sizeof(vis));
  memset(pre,-1,sizeof(pre));
  while(!q.empty()){
    int now = q.front();  q.pop();
    if(vis[now] == 2)continue;
    vis[now] = 2;
    if(now == si)break;
    FOR(i,N+1){
      int next = i;
      if(vis[next])continue;
      if(m[now][next]>0)
        vis[next]=1, pre[next]=now, q.push(next);
  } }
  int nt = si, ret = (1<<30);
  while(pre[nt] != -1){
    int bf = pre[nt];
    ret = min(ret,m[bf][nt]);  nt = bf;
  }
  if(ret >= (1<<30))return 0;
  nt = si;
  while(pre[nt] != -1){
    int bf = pre[nt];
    m[bf][nt] -= ret;  m[nt][bf] += ret;  nt = bf;
  }
  return ret;
}
```

```
int main(){
  int ntc;
  scanf("%d",&ntc);
  while(ntc--){
    scanf("%d%d",&N,&M);
    memset(d,0,sizeof(d));
    memset(m,0,sizeof(m));
    FOR(i,M){
      int a,b,c; scanf("%d%d%d",&a,&b,&c);
      d[a]++;  d[b]--;
      if(!c) m[a][b]++;
    }
    int can = 1;
    for(int i=1; can && i<=N; i++){
      int x = abs(d[i]);  d[i]>>=1;
      if(x&1) can=0;
    }
    if(can){
      int tflow = 0;
      REP(i,1,N){
        if(d[i]<0)m[i][N+1] = -d[i];
        else if(d[i]>0)m[0][i]=d[i], tflow+=d[i];
      }
      int tans, ans = 0;
      while((tans=flow(0,N+1)) > 0)ans += tans;
      can = can && (ans == tflow);
    }
    puts(can ? "possible" : "impossible");
} }
```

3.13 Max Bipartite Matching

```
vi match, vis;

int aug(int l){
  if(vis[l])return 0;
  vis[l] = 1;
  for(auto r: AdjList[l])
    if(match[r]==-1|| aug(match[r])){
      match[r]= l; return 1;
    }
  return 0;
}
```

3.14 Minimum Vertex Cover

```
vi AdjList[100010];  int memo[100010][2];

int solve(int idx, int flag, int prev){
  int ans = 0;
  if(memo[idx][flag] != -1) return memo[idx][flag];
  if(AdjList[idx].size()==1 && \
      AdjList[idx][0]==prev)
    return memo[idx][flag] = flag;
  //reccurence
  if(flag == 0){
    ans = 0;
    for(auto j: AdjList[idx]) if(prev != j)
      ans += solve(j,1,idx);
  }
  if(flag == 1){
    ans=1;
    for(auto j: AdjList[idx]) if(prev != j)
      ans += min(solve(AdjList[idx][j],1,idx), \
            solve(AdjList[idx][j],0,idx));
  }
  return memo[idx][flag] = ans;
}
int main(){
  int n; scanf("%d",&n);
  FOR(i,n-1){
    int a,b; scanf("%d %d",&a,&b);
    AdjList[a].pb(b);  AdjList[b].pb(a);
  }
  memset(memo,-1,sizeof(memo));
  int ans = solve(1,0,-1);
  memset(memo,-1,sizeof(memo));
  int ans2 = solve(1,1,-1);
  printf("%d\n", min(ans,ans2));
}
```

3.15 Articulation Point and Bridge

```
vi dfs_num, dfs_low, dfs_parent;
int dfs_counter = 0, rootChildren, dfs_root;

void articulationPoint(int u){
  dfs_num[u] = dfs_low[u] = dfs_counter++;
  for(auto v: AdjList[u]){
    if(dfs_num[v] == INF){
      dfs_parent[v] = u;
      if(u == dfsRoot)rootChildren++;
      articulationPoint(v);
      // Arti. Point
      if(dfs_low[v] >= dfs_num[u])
        flag[u] = true;
      // bridge Edge(u,v)
      if(dfs_low[v] > dfs_num[u])
        dfs_low[u] = min(dfs_low[u],dfs_low[v]);
    } else if(v != parent[u]){
      dfs_low[u] = min(dfs_low[u],dfs_num[v]);
} } }
//inside int main
dfs_counter = 0;  dfs_num.assign(V,INF);
dfs_low.assign(V,0);  dfs_parent(V,0);
memset(flag,false,sizeof(flag));
FOR(i,V) if(dfs_num[i] == INF){
  dfs_root = i;  rootChildren = 0;
  articulationPoint(i);
  flag[i] = (rootChildren>1);
}
```

3.16 SCC - Tarjan Algorithm

```
int cnt = 0;  vi num, low, isSt;  stack<int> st;

int tarjan(int u){
  if(num[u] != -1) return !isSt[u] ? INF : num[u];
  // reccurence
  num[u] = low[u] = ++cnt;
  st.push(u);  isSt[u] = 1;
  for(int v: adjList[u]) // move
    low[u] = min(low[u], tarjan(v));
  if(num[u] == low[u]){ // pop, if root
    while(st.top() != u){
      low[st.top()] = low[u];
      isSt[st.top()] = 0;  st.pop();
    }
    isSt[u] = 0;  st.pop();
  }
  return low[u];
}
```

3.17 SCC - Kosaraju Algorithm

```
// dfs normally, st.push(u) at the end
REP(i,1,n) dfs(i,0);
// reverse the direction of edges
// dfs from the stack, scc[u] = cnt at the end
while( !st.empty() ){
  int u=st.top();  st.pop();  cnt++;  dfs(u,1);
}
```

3.18 Hopcroft Karp

```
struct graph{
  int L, R;
  vector<vi> adj;
  graph(int L, int R) : L(L), R(R), adj(L+R) {}
  void addEdge(int u, int v){
    adj[u].pb(v+L);  adj[v+L].pb(u);
  }
  int maximum_matching(){
    vi level(L), mate(L+R,-1);

    function<bool(void)> levelize = [&]() { // BFS
      queue<int> Q;
      FOR(u,L){
        level[u] = -1;
        if(mate[u] < 0)level[u] = 0, Q.push(u);
      }
      while(!Q.empty()){
```

```
        int u = Q.front();  Q.pop();
        for(int w: adj[u]){
          int v = mate[w];
          if(v < 0)return true;
          if(level[v] < 0){
            level[v] = level[u]+1;
            Q.push(v);
      }   }   }
      return false;
    };
    function<bool(int)> augment = [&](int u) {
      for(int w: adj[u]){ // DFS
        int v = mate[w];
        if(v<0 || (level[v]>level[u]&&augment(v))){
          mate[u] = w;  mate[w] = u;  return true;
      } }
      return false;
    };
    int match = 0;
    while(levelize()) FOR(u,L)
      if (mate[u]<0 && augment(u))match++;
    return match;
} };
int main(){
  int L, R, m, u, v;
  scanf("%d %d %d",&L,&R,&m);  graph g(L,R);
  FOR(i,m)scanf("%d %d",&u,&v), g.addEdge(u,v);
  printf("%d\n",g.maximum_matching());
}
```

3.19 Euler Path + Euler Tour

```
#define all(c) ((c).begin()), ((c).end())

struct graph{
  int n;
  struct edge{ int src, dst, rev; };
  vector< vector<edge> > adj;
  graph() : n(0) {}
  //graph(int n) : n(n), adj(n) {}
  void addEdge(int src, int dst){
    n = max(n,max(src,dst)+1);
    if(adj.size() < n)adj.resize(n);
    adj[src].pb({src,dst,(int)adj[dst].size()});
    adj[dst].pb({dst,src,(int)adj[src].size()-1});
  }
  // destructive
  vi path;
  void visit(int u){
    while(!adj[u].empty()){
      auto e = adj[u].back();
      adj[u].pob();
      if(e.src >= 0){
        adj[e.dst][e.rev].src = -1;  visit(e.dst);
    } }
    path.pb(u);
  }
  vi eulerianPath(){
    int m = 0, s = -1;
    FOR(u,n){
      m += adj[u].size();
      if(adj[u].size()%2 == 1)s = u;
    }
    path.clear();  if(s >= 0)visit(s);
    if(path.size() != m/2+1)return {};
    return path;
  }
  vi eulerianTour(){
    int m = 0, s = 0;
    FOR(u,n){
      m += adj[u].size();
      if(adj[u].size() > 0)s = u;
    }
    path.clear();  visit(s);
    if(path.size()!=m/2+1 || path[0]!=path.back())
      return {};
    return path;
} };
int main(){
  int ncase;
  scanf("%d", &ncase);
  REP(icase,1,ncase){
    if(icase > 1) printf("\n");
    printf("Case #%d\n",icase);
    int m;  graph g;  scanf("%d",&m);
    FOR(i,m){
      int s,t; scanf("%d %d", &s, &t);
      g.addEdge(s-1,t-1);
    }
    auto path = g.eulerianTour();
    if(path.empty())printf("may be lost\n");
    else FOR(i,path.size()-1){
      printf("%d %d\n", path[i]+1, path[i+1]+1);
} } }
```

3.20 Edmonds Blossom

```
// Given a graph G = (V,E), a matching M in G is a
// set of pairwise non-adjacent edges; that is, no
// two edges share a common vertex. A maximum
// matching is a matching that contains the largest
// possible number of edges. There may be many
// maximum matchings. The matching number of a
// graph is the size of a maximum matching. The
// problem is to find maximum matching of for given
// graph.
int match[150], V = 4, parent[150], base[150];
bool vis[150], temp[150], blossom[150];
vi vec[150];  queue<int> q;

int LCA(int a,int b){
  memset(temp,false,sizeof(temp));
  while(1){
    a = base[a];  temp[a] = true;
    if(match[a] == -1)break;
    a = parent[match[a]];
  }
  while(1){
    b = base[b];
    if(temp[b] == true)return b;
    b = parent[match[b]];
} }
void markPath(int v, int b, int children){
  for(; base[v]!=b; v=parent[match[v]]){
    blossom[base[v]] = true;
    blossom[base[match[v]]] = true;
    parent[v] = children;  children = match[v];
} }
int findPath(int root){
  memset(vis,false,sizeof(vis));
  memset(parent,-1,sizeof(parent));
  FOR(a,V)base[a] = a;  vis[root] = true;
  while(!q.empty())q.pop();  q.push(root);
  while(!q.empty()){
    int pos = q.front();  q.pop();
    int sz = vec[pos].size();
    FOR(a,sz){
      int nx = vec[pos][a];
      if(base[nx]==base[pos] || match[pos]==nx)
        continue;
      if(nx==root || match[nx]!=-1 && \
          parent[match[nx]]!=-1){
        int curbase = LCA(pos,nx);
        memset(blossom,false,sizeof(blossom));
        markPath(pos,curbase,nx);
        markPath(nx,curbase,pos);
        FOR(b,V) if(blossom[base[b]] == true){
          base[b] = curbase;
          if(vis[b]==false){
            vis[b] = true;  q.push(b);
      } } }
      else if(parent[nx]==-1){
        parent[nx] = pos;
        if(match[nx] == -1) return nx;
        nx = match[nx];  vis[nx] = true;
        q.push(nx);
  } } }
  return -1;
}
int edmonds(){
  int total = 0;  memset(match,-1,sizeof(match));
  FOR(a,V) if(match[a]==-1){
    int nx = findPath(a);
    while(nx!=-1){
```

```
      int pnx = parent[nx];  int ppnx = match[pnx];
      match[nx] = pnx;  match[pnx] = nx;
      nx = ppnx;
  } }
  FOR(a,V) if(match[a] != -1) total++;
  return total/2;
}
```

3.21 Aho Corasick

```
struct Trie{
  Trie *next[52], *fail;  vi label;  bool match;
  Trie(){
    fail = NULL;  match = false;  label.clear();
    memset(next,0,sizeof(next));
} };
void build_Trie(Trie** a, int val){
  int len = strlen(q_in);  Trie *p = *a;
  FOR(w,len){
    int idx = convert(q_in[w]);
    if(p->next[idx] == NULL)
      p->next[idx] = new Trie();
    p = p->next[idx];
  }
  p->label.pb(val);
}
void build_Automation(Trie* root){
  queue<Trie*> q;  Trie *state, *fail;
  root->fail = NULL;  q.push(root);
  while(!q.empty()){
    state = q.front();  q.pop();
    FOR(a,52){
      if(state->next[a] == NULL) continue;
      q.push(state->next[a]);
      fail = state->fail;
      while(fail!=NULL && fail->next[a]==NULL)
        fail = fail->fail;
      if(fail == NULL)state->next[a]->fail = root;
      else state->next[a]->fail = fail->next[a];
} } }
void search_pattern(Trie* root){
  Trie *p = root, *check;
  int len = strlen(main_in);
  FOR(a,len){
    int nx = convert(main_in[a]);
    while(p!=root && p->next[nx]==NULL)
      p = p->fail;
    p = p->next[nx];
    if(p == NULL)p = root;
    check = p;
    while(check!=root && check->match==false){
      int sz = check->label.size();
      FOR(b,sz)mark[check->label[b]] = true;
      check->match = true;  check = check->fail;
} } }
void clear_Automation(Trie* root){
  queue<Trie*> q;  Trie *nx;  q.push(root);
  while(!q.empty()){
    nx = q.front();  q.pop();
    FOR(a,52) if(nx->next[a] != NULL)
      q.push(nx->next[a]);
    nx->label.clear();  delete nx;
} }
```

3.22 Maximum Clique

```
// max complete subgraph in a Graph
class MaxClique{
public:
  static const int MV = 210;
  int V, el[MV][MV/30+1], dp[MV];
  int ans, s[MV][MV/30+1];  vi sol;

  void init(int v){
    V = v;  ans = 0;
    memset(el,0,sizeof(el));
    memset(dp,0,sizeof(dp));
  }
  /* Zero Base */
  void addEdge(int u, int v){
    if(u > v)swap(u, v);
    if(u == v)return;
    el[u][v/32] |= (1<<(v%32));
  }
  bool dfs(int v, int k){
    int c = 0, d = 0;
    FOR(i,((V+31)/32)){
      s[k][i] = el[v][i];
      if(k != 1)s[k][i] &= s[k-1][i];
      c += __builtin_popcount(s[k][i]);
    }
    if(c == 0){
      if(k > ans)
        ans = k, sol.clear(), sol.pb(v), return 1;
      return 0;
    }
    FOR(i,((V+31)/32)) for(int a=s[k][i]; a; d++){
      if(k+(c-d) <= ans)return 0;
      int lb = a&(-a), lg = 0;  a ^= lb;
      while(lb != 1)
        lb = (unsigned int)(lb)>>1, lg++;
      int u = i*32 + lg;
      if(k+dp[u] <= ans) return 0;
      if(dfs(u,k+1)) sol.pb(v), return 1;
    }
    return 0;
  }
  int solve(){
    FORD(i,V) dfs(i,1), dp[i] = ans;
    return ans;
} };
```

3.23 DSU on Tree

```
void add(int u, int p, int val){
  cnt[col[u]] += val;
  for(auto v: adjList[u]){
    if(v==p || big[v]) continue;
    add(v,u,val);
  }
}
void dsu(int u, int p, bool keep){
  int maks = -1, bigChild = -1;
  for(auto v: adjList[u]){
    if(v == p) continue;
    if(maks< sz[v]) maks = sz[v], bigChild = v;
  }
  for(auto v: adjList[u]){
    if(v == p || v == bigChild) continue;
    dsu(v,u,0);
  }
  if(bigChild != -1){
    dsu(bigChild,u,1);
    big[bigChild] = true;
  }
  add(u,p,1);
  if(bigChild != -1) big[bigChild] = false;
  if(keep == 0) add(u,p,-1);
}
```

3.24 Recover Min-Vertex Cover from Maximum Matching

Suppose you have left and right parts in bipartite graph and you found the maximum matching M. Let's define orientation of edges. Those edges that belong to M will go from right to left, all other edges will go from left to right. Now run DFS starting at all left vertices that are not incident to edges in M. Some vertices of the graph will become visited during this DFS and some not-visited. To get minimum vertex cover you take all visited right vertices of M, and all not-visited left vertices of M.

4 String Processing

4.1 KMP Algorithm

```
char s[MAX+5], a[MAX+5];  int pos[MAX+5];

int KMP(int k){
  int la = strlen(a), ls = strlen(s), temp, res;
  temp = pos[0] = -1;
  FOR(i,la){
    while(temp>=0 && a[i]!=a[temp])
      temp = pos[temp];
    pos[i+1] = ++temp;
  }
  temp = res = 0;
  FOR(i,ls){
    while(temp>=0 && s[i]!=a[temp])
      temp = pos[temp];
    if(++temp == la){ temp=pos[temp]; return 1; }
  }
  return 0; //not found
}
```

4.2 Suffix Array

```
int dat[100005][25];  pair<pii,int> ret[100005];

void sparse(char* s, int n){
  FOR(i,100003) FOR(j,23)dat[i][j] = -1;
  FOR(j,20){
    //copy data
    ret[0] = mp(mp(-1,-1),-1);
    FOR(i,n){
      ret[i+1].se = i;
      if(j==0)ret[i+1].fi = mp((int)s[i],-1);
      else{
        int mask = (i+(1<<(j-1)))%n;
        ret[i+1].fi.fi = dat[i][j-1];
        ret[i+1].fi.se = dat[mask][j-1];
    }   }
    sort(ret+1,ret+1+n);
    //iterate
    int pos=0;
    REP(i,1,n){
      if(ret[i-1].fi == ret[i].fi)
        dat[ret[i].se][j] = pos;
      else dat[ret[i].se][j] = ++pos;
} } }
int main(){
  int t,n;  char s[200005];
  scanf("%d",&t);
  while(t--){
    scanf("%d %s",&n,s);  sparse(s,n);
    int res = 0;
    FOR(i,n){
      //compare with pos
      int posa = i, posr = res;
      FORD(j,20) if(n&(1<<j)){
        if(dat[posa][j] == dat[posr][j]){
          posa = (posa+(1<<j))%n;
          posr = (posr+(1<<j))%n;
          continue;
        }
        if(dat[posa][j] < dat[posr][j])res = i;
        break;
    } }
    printf("%d\n",res);
} }
```

```
//SA Hansvin
int n, sparse[21][5*N+5];
int bucket[5*N+5]; char s[5*N+5];
pair<pii,int> lst[5*N+5], lst2[5*N+5];
void radix(){
  int nn = max(n,10);
  //second
  memset(bucket,0,sizeof(bucket));
  FOR(i,n) bucket[lst[i].fi.se]++;
  REP(i,1,nn-1) bucket[i] += bucket[i-1];
  FORD(i,n){
    int pos = bucket[lst[i].fi.se]-1;
    bucket[lst[i].fi.se]--;
    lst2[pos] = lst[i];
  }
  //first
  memset(bucket,0,sizeof(bucket));
  FOR(i,n) bucket[lst2[i].fi.fi]++;
  REP(i,1,nn-1) bucket[i]+=bucket[i-1];
  FORD(i,n){
    int pos = bucket[lst2[i].fi.fi]-1;
    bucket[lst2[i].fi.fi]--;
    lst[pos] = lst2[i];
} }
void build(){
  FOR(i,20){
    FOR(j,n){
      lst[j].se = j;
      if(i==0)lst[j].fi = mp(s[j]-'0',0);
      else lst[j].fi = mp(sparse[i-1][j], \
           sparse[i-1][(j+(1<<(i-1)))%n]);
    }
    radix();
    int pos = 0;
    sparse[i][lst[0].se] = 0;
    REP(j,1,n-1) sparse[i][lst[j].se] = \
      (lst[j].fi==lst[j-1].fi) ? pos:++pos;
} }
int main(){
  scanf("%d",&n); scanf("%s", s);
  build(); int res=0;
  FOR(i,n){
    int posa = i, posr = res;
    FOR(j,20) if(n&(1<<j)){
      if(sparse[j][posr] == sparse[j][posa]){
        posa = (posa+(1<<j))%n;
        posr = (posr+(1<<j))%n;
        continue;
      }
      if(sparse[j][posr] >sparse[j][posa]) res=i;
      break;
  } }
  printf("%d\n", res);
}
```

4.3 Enhanced Suffix Array

```
#define MAX_N 100000
typedef struct{ int a[2], idx; } ttemp;
bool cmp(const ttemp& a, const ttemp& b){
  return (a.a[0]==b.a[0]) ? (a.a[1]<b.a[1]) : \
    (a.a[0]<b.a[0]);
}
ttemp L[MAX_N], temp[MAX_N];
int step, p[20][MAX_N], lt, lq, a[MAX_N+1];

void rasort(ttemp x[], int n){
  int k = -1;  memset(a,0,sizeof(a));
  FOR(i,n)a[x[i].a[1]+1]++, k = max(k,x[i].a[1]+1);
  REP(i,1,k)a[i] += a[i-1];
  FORD(i,n)temp[ --a[x[i].a[1]+1] ] = x[i];
  memset(a,0,sizeof(a));  k = -1;
  FOR(i,n){
    a[temp[i].a[0]+1]++;
    k = max(k, temp[i].a[0]+1);
  }
  REP(i,1,k)a[i] += a[i-1];
  FORD(i,n)x[ --a[temp[i].a[0]+1] ] = temp[i];
}
void generate(char s[], int p[][MAX_N]){
  int cnt;  FOR(i,lt)p[0][i] = s[i];
  for(step=1,cnt=1; (cnt>>1)<lt; step++,cnt<<=1){
    FOR(i,lt){
      L[i].a[0] = p[step-1][i];  L[i].idx = I;
      L[i].a[1] = (i+cnt<lt)?p[step-1][i+cnt]:-1;
    }
    rasort(L,lt);  //sort(L,L+lt); => n log^2 n
    FOR(i,lt) p[step][L[i].idx] = (i && L[i].a[0] \
      == L[i-1].a[0] && L[i].a[1] == L[i-1].a[1]) \
      ? p[step][L[i-1].idx] : i;
  }
  step--;
}
char pattern[1100], text[100010];
//compare pattern suffix[idx]
inline int compare(int idx){
  int i = 0;
  for(;i<lq && idx<lt; i++,idx++){
    if(pattern[i] > text[idx])return -1;
```

```
    if(pattern[i] < text[idx])return 1;
  }
  return (i == lq) ? 0 : -1;
}
inline int lcp(int x, int y){
  if(x == y)return lt-x;
  int res = 0;
  for(int k = step-1; k>=0 && x<lt && y<lt; k--)
    if (p[k][x]==p[k][y])
      x+=(1<<k),y+=(1<<k),res+=(1<<k);
  return res;
}
//cari many pattern pada long text, preprocess text
int L = 0, R = lt-1, mid, found = false;
while(L <= R && !found){
  mid = (L+R)>>1;  int temp = compare(arr[mid]);
  if(temp==0)found = 1;
  else if(temp==1)R = mid-1;
  else L = mid+1;
}
//Longest palindrome
S = S + rev(S), edit LCP
for(int i=0;i<lt/2-1 ;i++){
  res >?= lcp(lt-i-1,i+1);
  if(i)res >?= 1 + lcp(lt-i,i+1);
}
//Distinct substring
if(lt&1)res = ((lt+1)>>1)*lt;
else res = (lt>>1) * (lt+1);
FOR(i,lt)arr[p[step-1][i]] = i;
FOR(i,lt-1)res -= lcp(arr[i],arr[i+1]);
//longest palindrome, maximum K change
int longpal(int left, int right, int k){
  int temp = 0, a;
  for(int j=0; j<=k; ){
    a = lcp(right,left);
    temp += 2*a;  left +=a;  right+=a;
    if(a==0){
      if(j==k)break;
      temp+=2;  left++;  right++;  j++;
    }
    if(left>= lt || right >= lt/2)break;
  }
  return temp;
}
```

4.4 Edit Distance

```
int dp[MAXN][MAXN];  char s1[MAXN], s2[MAXN];

int main(){
  int len1,len2;
  while(scanf("%d",&len1)!=EOF){
    if(len1>0) scanf("%s",s1);
    scanf("%d",&len2);
    if(len2>0) scanf("%s", s2);
    bool one = (len1==0), two = (len2==0);
    FILL(dp,0);
    dp[0][0] = (s1[0]!=s2[0]);
    REP(i,1,len1-1)dp[i][0] = (s1[i]==s2[0]) ? \
      i : min(i+1,dp[i-1][0]+1);
    REP(i,1,len2-1) dp[0][i] = (s1[0]==s2[i]) ? \
      i : min(i+1,dp[0][i-1]+1);
    REP(i,1,len1-1) REP(j,1,len2-1){
      dp[i][j] = INF;
      if(s1[i] == s2[j])dp[i][j] <?= dp[i-1][j-1];
      dp[i][j] = min(dp[i][j],min(dp[i-1][j]+1, \
        min(dp[i][j-1]+1,dp[i-1][j-1]+1)));
    }
    printf("%d\n",(one^two) ? (max(len1,len2)) : \
      dp[len1-1][len2-1]);
} }
```

4.5 kth Permutation

```
string kthpermute(ll k, string s){
  int len = s.length();  char ss[len+2];
  ll fac = 1;
  sprintf(ss,"%s",s.c_str());
  REP(i,2,len-1)fac = (ll)i * (ll)fac;
  FOR(i,len-1){
    int tidx=(k/fac)%(len-i);  char ch=ss[i+tidx];
    REV(j,i+tidx,i+1)ss[j] = ss[j-1];
    ss[i] = ch;  fac = fac/(ll)(len-i-1);
  }
  return (s = ss);
}
int main(){
  string s = "abcdefghijklmnopqrst";
  string res = kthpermute(1234567654321, s);
  printf("%s\n", res.c_str());
}
```

4.6 Manacher's Algorithm (Longest Palindromic Substring)

```
int lps[200005];  char s[100005];
void manacher(int n){
  FOR(i,200003) lps[i]=0;  lps[0]=0;  lps[1]=1;
  REP(i,2,2*n){
    int l=i/2, r=(i-1)/2;
    l-=lps[i]/2; r+=lps[i]/2;
    // widen
    while(l!=0&& r!=n-1&&s[l-1]==s[r+1]) l--, r++;
    lps[i]= r-l+1;
    if(lps[i]<=1) continue;
    // jump
    int best = i-1;
    while(lps[best]-best< lps[i]-i)best--;
    for(int j=i-1, k=i+1; j>best; j--,k++)
      lps[k] = lps[j];
    lps[i+i-best] = min(lps[best],lps[i]-(i-best));
    i = i+i-best-1;
} }
```

5 ETC

5.1 Soal TIMO 2010

```
const ll mod = 1000000007;
ll  nck[100][100];  ll  memo[10][100];
int arr[10];

ll f(int x, int m){
  if(m ==  0)return 1;
  if(x == 10)return 0;
  if(memo[x][m] != -1)return memo[x][m];
  ll& ret = memo[x][m] = 0;
  FOR(i,min(m,arr[x]))
    ret += f(x+1,m-i)*nck[m][i] % mod;
  return ret;
}
int main(){
  nck[0][0] = 1;
  REP(i,1,99){
    nck[i][0] = nck[i][i] = 1;
    REP(j,1,i-1)nck[i][j]=(nck[i-1][j]+nck[i-1] \
    [j-1])%mod;
  }
  int T;
  scanf("%d",&T);
  while(T--){
    REP(i,1,9)scanf("%d",&arr[i]);
    int total = accumulate(arr+1,arr+10,0);
    ll ans = 0;
    REP(k,1,9) if(arr[k] != 0){
      arr[k]--;  memset(memo,-1,sizeof(memo));
      ll p = k;
      REP(len,1,total){
        ans = (ans+(f(1,len-1)*p)%mod) % mod;
        p   = (p*10+k) % mod;
      }
      arr[k]++;
    }
    printf("%I64d\n", ans );
} }
// Problem statement: Given the number of available
// digits of digit 1 to 9, sum all possible numbers
// that can be generated by those available digits.
```

5.2 Meet in the Middle

```
long sumsLowerBound(long[] A, long B){
  int n = A.length;
  int sizeL = 1<<(n/2), sizeR = 1<<(n-n/2);
  long[] SumsL = new long[sizeL];
  long[] SumsR = new long[sizeR];
  FOR(i,sizeL) FOR(j,n/2) if((i&(1<<j)) > 0)
    SumsL[i] += A[j];
  FOR(i,sizeR) FOR(j,n-n/2) if((i&(1<<j)) > 0)
    SumsR[i] += A[j+n/2];
  Arrays.sort(SumsL);  Arrays.sort(SumsR);
  int PL = 0, PR = sizeR-1;
  long cur=Long.MIN_VALUE;
  while(PL<sizeL && PR >= 0){
    if(SumsL[PL]+SumsR[PR] <= B)
      cur = Math.max(cur,SumsL[PL++]+SumsR[PR]);
    else PR--;
  }
  return cur;
}
```

5.3 Huffman Code

```
#include <fstream>
int main(){
  priority_queue< ll,vector<ll>,greater<ll> > q;
  ifstream in("plank.in");
  ofstream out("plank.out");
  int N;  ll cost = 0;  in >> N;
  FOR(i,N){ ll x;  in >> x;  q.push(x); }
  for(; (int)q.size() >= 2;){
    ll a = q.top();  q.pop();
    ll b = q.top();  q.pop();
    cost += a+b;  q.push(a + b);
  }
  out << cost << "\n";
}
```

5.4 Hungarian Algorithm

```
int cost[64][64], lvisit[64], rvisit[64];
int lvalue[64], rvalue[64], rmatch[64], n;

bool findmatch(int s){
  lvisit[s] = true;
  FOR(i,n){
    if(rvisit[i]||lvalue[s]+rvalue[i]!=cost[s][i])
      continue;
    rvisit[i] = true;
    if(rmatch[i]==-1 || findmatch(rmatch[i])){
      rmatch[i] = s;  return true;
  } }
  return false;
}
void normalize(){
  int down = INF;
  FOR(i,n) if(lvisit[i]) FOR(j,n) if(!rvisit[j])
    down = min(down, \
      lvalue[i]+rvalue[j]-cost[i][j]);
  FOR(i,n){
    if(lvisit[i])lvalue[i] -= down;
    if(rvisit[i])rvalue[i] += down;
} }
int calc(){
  FOR(i,n){
    lvalue[i] = *max_element(cost[i],cost[i]+n);
    rvalue[i] = 0;
  }
  FOR(i,n)rmatch[i] = -1;
  FOR(i,n){
    FILL(lvisit,0);  FILL(rvisit,0);
    if(findmatch(i))continue;
    i--;  normalize();
  }
  int ret = 0;
  FOR(i,n)ret += cost[rmatch[i]][i];
  return ret;
}
```

5.5 Stable Marriage

```
int m,n;
int L[max_man][max_woman],R[max_woman][max_man];
int L2R[max_man],R2L[max_woman], p[max_man];

void stableMarriage(){
  static int p[128];
  memset(R2L,-1,sizeof(R2L));
  memset(p,0,sizeof(p));
  FOR(i,m) for(int man=i,hubby; man>=0; man=hubby){
    int wom;
    while(1){
      wom = L[man][p[man]++];
      if(R2L[wom]<0 || R[wom][man] > \
        R[wom][R2L[wom]]) break;
    }
    hubby = R2L[wom];  R2L[L2R[man]=wom] = man;
} }
```

5.6 CSOD

```
ll sum(ll a, ll b){
  return (b*(b+1)>>1) - ((a-1)*a>>1);
}
ll CSOD(LL n){
  ll i, m = n>>1, out2 = 0, j = sqrt(n);
  REP(i,2,j) out2=(n/i)*i;
  for(i--;j>=1;j--)
    out2 += (j*sum(i+1,n/j)), i=(n/j);
  return out2 -= ((n*(n+1)>>1)-1);
}
```

5.7 RDN (goto 0001/01/01)

```
// Rata Die day one is 0001-01-01
int rdn(int y, int m, int d){
  if(m < 3) y--, m += 12;
  int res = 365*y + y/4 - y/100 + y/400;
  return res + (153*m-457)/5 + d-306;
}
```

6 Mathematics

6.1 Big Integer

```
const int maxlength=1000;
class bigint{
public:
  // 1. Basic
  int oper,length,a[maxlength];
  bigint(int=0);
  ~bigint();
  void check();
  void operator=(bigint m);
  void operator=(int m);
  void operator=(char *s);
  int max(int a,int b);
  // 2. Relational operations
  bool operator<(bigint m);
  bool operator<=(bigint m);
  bool operator>(bigint m);
  bool operator>=(bigint m);
  bool operator==(bigint m);
  bool operator!=(bigint m);
  // 3. Arithmetic operations
  bigint operator-();
  bigint operator+(bigint m);
  void operator+=(bigint m);
  bigint operator-(bigint m);
  void operator-=(bigint m);
  bigint operator*(bigint m);
  bigint operator*(int m);
  void operator*=(bigint m);
  void operator*=(int m);
  bigint operator/(bigint m);
  bigint operator/(int m);
  void operator/=(bigint m);
  void operator/=(int m);
  bigint operator%(bigint m);
```

```
  bigint operator%(int m);
  void operator%=(bigint m);
  void operator%=(int m);
};
  // 4. Input/ Output
bool read(bigint &m);
void write(bigint m);
void swrite(char *s,bigint m);
void writeln(bigint m);
  // 5. Mathematical functions
bigint abs(bigint m);
bigint sqr(bigint m);
bigint sqrt(bigint m);
bigint gcd(bigint a,bigint b);
bigint lcm(bigint a,bigint b);
bigint max(bigint a,bigint b);
bigint min(bigint a,bigint b);
bigint fpow(bigint a,bigint b);

// B. Class Implementation
// 1. Basic
bigint::bigint(int v){
  (*this) = v;
  this->check();
}
bigint::~bigint(){}
  void bigint::check(){
    for(; length>0 && a[length]==0; length--);
    if(length==0)oper=1;
}
void bigint::operator=(bigint m){
  oper = m.oper;  length = m.length;
  memcpy(a,m.a,maxlength*sizeof(int));
  this->check();
}
void bigint::operator=(int m){
  oper = (m>0)?1:-1;  m = abs(m);
  memset(a,0,maxlength*sizeof(int));
  for(length=0; m>0; m/=10000)a[++length] = m%10000;
  this->check();
}
void bigint::operator=(char *s){
  int i,L;  (*this)=0;
  if(s[0]=='-' || s[0]=='+'){
    if(s[0] == '-')oper=-1;
    L = strlen(s);  FOR(i,L)s[i] = s[i+1];
  }
  L = strlen(s);  length = (L+3)/4;
  FOR(i,L)a[(L-i+3)/4] = a[(L-i+3)/4]*10+(s[i]-48);
  this->check();
}
int bigint::max(int a,int b){return (a>b)?a:b;}

// 2. Relational operations
bool bigint::operator<(bigint m){
  if(oper!=m.oper)return oper<m.oper;
  if(length!=m.length)
    return oper*length<m.length*oper;
  REV(i,length,1) if(a[i] != m.a[i])
    return a[i]*oper < m.a[i]*oper;
  return false;
}
bool bigint::operator<=(bigint m){return !(m<(*this));
bool bigint::operator>(bigint m){ return m<(*this);}
bool bigint::operator>=(bigint m){return !((*this)<m);
bool bigint::operator==(bigint m){
  return (!((*this)<m)) && (!(m<(*this))); }
bool bigint::operator!=(bigint m){
  return ((*this)<m) || (m<(*this)); }

// 3. Arithmetic Operations
bigint bigint::operator-(){
  bigint c = (*this);  c.oper = -c.oper;
  c.check();  return c;
}
bigint bigint::operator+(bigint m){
  if(m.length == 0)return (*this);
  if(length == 0)return m;
  if(oper == m.oper){
    bigint c;  c.oper=oper;
    c.length = max(length,m.length)+1;
    for(int i=1,temp=0; i<=c.length; i++)
      c.a[i] = (temp=(temp/10000+a[i]+m.a[i]))%10000;
    c.check();  return c;
  }
  return (*this)-(-m);
}
bigint bigint::operator-(bigint m){
  if(m.length == 0)return (*this);
  if(length == 0)return (-m);
  if(oper == m.oper){
    bigint c;
    if(abs(*this)>=abs(m)){
      c.oper = oper;  c.length=length;
      for(int i=1,temp=0; i<=length; i++)
        c.a[i]=((temp=(-int(temp<0)+a[i]-m.a[i])) \
        +10000)%10000;
      c.check();
      return c;
    }
    return -(m-(*this));
  }
  return (*this)+(-m);
}
bigint bigint::operator*(bigint m){
  bigint c;  c.oper = oper*m.oper;
  c.length = length+m.length;
  REP(i,1,m.length){
    int number = m.a[i], j, temp = 0;
    REP(j,1,length)c.a[i+j-1] += number*a[j];
    if(i%10==0 || i==m.length) REP(j,1,c.length)
      c.a[j] = (temp=(temp/10000)+c.a[j]) % 10000;
  }
  c.check();  return c;
}
bigint bigint::operator*(int m){
  if(m<0)return -((*this)*(-m));
  if(m>100000)return (*this)*bigint(m);
  bigint c;  c.length=length+2; c.oper=oper; int t=0;
  REP(i,1,c.length)c.a[i]=(t=(t/10000+a[i]*m))%10000;
  c.check();  return c;
}
bigint bigint::operator/(bigint m){
  if(m.length==0){
    printf("Division by zero.\n"); exit(0);
  }
  if(abs(*this)<abs(m))return 0;
  bigint c, left; c.oper = oper/m.oper;  m.oper = 1;
  c.length = length-m.length+1;
  left.length = m.length-1;
  memcpy(left.a+1,a+length-left.length+1, \
    left.length*sizeof(int));
  REV(i,c.length,1){
    left = left*10000+a[i];
    int head = 0, tail = 10000, mid;
    while(head+1<tail){
      mid = (head+tail)/2;
      if(m*mid <= left)head = mid;
      else tail = mid;
    }
    c.a[i] = head;  left -= m*head;
  }
  c.check();  return c;
}
bigint bigint::operator/(int m){
  if(m<0)return -((*this)/(-m));
  if(m>100000)return (*this)/bigint(m);
  bigint c;  c.oper=oper;  c.length=length;  int t=0;
  REV(i,c.length,1)c.a[i] = (t=(t%m*10000+a[i]))/m;
  c.check();  return c;
}
bigint bigint::operator %(bigint m){
  return (*this)-((*this)/m)*m; }
bigint bigint::operator%(int m){
  if(m<0)return -((*this)%(-m));
  if(m>100000)return (*this)%bigint(m);
  int t=0;
```

```
  REV(i,length,1)t = (t*10000+a[i]) % m;
  return t;
}
void bigint::operator+=(bigint m){ (*this)=(*this)+m;}
void bigint::operator-=(bigint m){ (*this)=(*this)-m;}
void bigint::operator*=(bigint m){ (*this)=(*this)*m;}
void bigint::operator/=(bigint m){ (*this)=(*this)/m;}
void bigint::operator%=(bigint m){ (*this)=(*this)%m;}
void bigint::operator*=(int m){ (*this) = (*this)*m; }
void bigint::operator/=(int m){ (*this) = (*this)/m; }
void bigint::operator%=(int m){ (*this) = (*this)%m; }

// 4. Input/Output
bool read(bigint &m){
  char s[maxlength*4+10];
  if(scanf("%s",&s)==-1)return false;
  for(int i=0;s[i];i++) if(!(s[i]>='0' && s[i]<='9' \
    || (s[i]=='+'||s[i]=='-') && i==0))return false;
  m=s;
  return true;
}
void swrite(char* s, bigint m){
  int L = 0;
  if(m.oper == -1)s[L++] = '-';
  sprintf(s+L,"%d",m.a[m.length]);
  for(; s[L]!=0; L++);
  REV(i,m.length-1,1)sprintf(s+L,"%04d",m.a[i]),L+=4;
  s[L] = 0;
}
void write(bigint m){
  if(m.oper == -1)printf("-");
  printf("%d",m.a[m.length]);
  REV(i,m.length-1,1)printf("%04d",m.a[i]);
}
void writeln(bigint m){ write(m);  printf("\n"); }

// 5. Mathematical functions
bigint sqr(bigint m){ return m*m; }
bigint sqrt(bigint m){
  if(m.oper<0 || m.length==0)return 0;
  bigint c, last, now, templast;
  c.length = (m.length+1)/2;
  c.a[c.length] = int(sqrt((double)m.a[c.length*2] \
    *10000+m.a[c.length*2-1])+1e-6);
  templast.length = c.length*2;
  templast.a[c.length*2-1] = (c.a[c.length]* \
    c.a[c.length])%10000;
  templast.a[c.length*2] = (c.a[c.length]* \
    c.a[c.length])/10000;
  templast.check();
  REV(i,c.length-1,1){
    last = templast;
    int head = 0, tail = 10000, mid, j, temp;
    while(head+1<tail){
      mid = (head+tail)/2;  now = last;
      now.a[2*i-1] += mid*mid;
      REP(j,i+1,c.length)now.a[i+j-1]+=mid*c.a[j]*2;
      now.length++;
      for (j=2*i-1,temp=0;j<=now.length;j++)
        now.a[j]=(temp=(temp/10000+now.a[j]))%10000;
      now.check();
      if(now<=m)templast = now, head=mid;
      else tail = mid;
    }
    c.a[i] = head;
  }
  c.check();  return c;
}
bigint abs(bigint m){
  bigint c = m;  c.oper=abs(c.oper);
  c.check();  return c;
}
bigint gcd(bigint a,bigint b){
  return b==0?a:gcd(b,a%b); }
bigint lcm(bigint a,bigint b){ return a*(b/gcd(a,b));}
bigint max(bigint a,bigint b){ return a>b?a:b; }
bigint min(bigint a,bigint b){ return a<b?a:b; }
bigint fpow(bigint a,bigint b){
  bigint zero = 0, one = 1, two = 2;
  if(b == zero)return one;
  if(b == one)return a;
  bigint half = fpow(a,b/two);
  if(b%two == 0)return half*half;
  else return half*half*a;
}
```

6.2 Fast Fourier Transform + NTT

```
// For NTT, let p = (2^k)*c + 1 be modulo prime
// need: size of polynom <= 2^k
// find prr (roots) => prr^1 .. prr^p-1 distinct
// dont forget divmod and other mods

typedef complex<double> cpx;
typedef vector<cpx> vec;
void FFT(vec& a, int sign = 1){
  int n = a.size();
  double theta = sign * TAU / n;
  // ll theta = fastExp(prr, (MOD-1)/n);
  // if(sign == -1) theta = fastExp(theta, MOD-2);
  for(int i=0,j=1; j<n-1; j++){
    for(int k=n>>1; k>(i^=k); k>>=1);
    if(j<i)swap(a[i],a[j]);
  }
  for(int m,mh=1; (m=mh<<1)<=n; mh=m){
    int irev = 0;
    for(int i=0; i<n; i+=m){
      cpx w = exp(cpx(0,theta*irev)); //e^(i*theta)
      // ll w = fastExp(theta, irev); pre-compute
      for(int k=n>>2; k>(irev^=k); k>>=1);
      for(int j=i; j<mh+i; j++){
        int k = j+mh;
        cpx x = a[j]-a[k];
        a[j] += a[k];
        a[k] = w*x;
  }   }   }
  if(sign == -1) FOR(i,n) a[i]/=n;
}
void polyMultiplication(vec& res, vec& a, vec& b){
  res.clear();  int n = a.size()+b.size()-1;
  REPP(i,1,n-1,i) if(n&i) n += i;
  a.resize(n);  b.resize(n);  FFT(a);  FFT(b);
  FOR(i,n)res.pb(a[i]*b[i]);  FFT(res,-1);
}
```

6.3 Gauss Elimination

```
typedef vector< vector<double> > Matrix;  Matrix M;

bool Gaussian () {
  int N = M.size();
  FOR(i,N){
    int idx = -1;
    REP(j,i,N-1) if(fabs(M[j][i])>EPS)
      idx = j, break;
    if(idx == -1){ return false; }
    REP(j,i,N)swap(M[i][j],M[idx][j]);
    REV(j,N,i)M[i][j] /= M[i][i];
    REP(j,i+1,N-1) REV(k,N,i)
      M[j][k] -= M[j][i]*M[i][k];
  }
  FORD(i,N+1) FOR(j,i) REV(k,N,i)
    M[j][k] -= M[j][i]*M[i][k];
  return true;
}
```

6.4 Number Theory

Bertrand's Ballot Theorem

Candidate A receives p votes and candidate B receives q votes,probability that A will be strictly ahead of B throughout count?

No Ties: $\frac{p-q}{p+q}$ Ties Allowed: $\frac{p+1-q}{p+1}$

Gambler's Ruin Theorem

Gambler's Ruin Theorem

Unfair: $P_1=\frac{1-\left(\frac{p}{q}\right)^{n_2}}{1-\left(\frac{p}{q}\right)^{n_1+n_2}}$ $P_2=\frac{1-\left(\frac{q}{p}\right)^{n_1}}{1-\left(\frac{q}{p}\right)^{n_1+n_2}}$

Fair: $P_1=\frac{n_2}{n_1+n_2}$ $P_2=\frac{n_1}{n_1+n_2}$

Pick Theorem

Let A : the area of a simply closed lattice polygon; B : the number of lattice points on the Polygon edges; I : the number of points in the interior of the polygon. Then **A = I+(1/2)B-1**

Forbenius Number

If GCD is 1 then required largest amount is (X*Y)-(X+Y) and total count is (X-1)*(Y-1)/2

Erdos Gallai Theorem

A sequence of non-negative integers $d_1 \leq \ldots \leq d_n$, can be represented as the degree sequence of a finite simple graph on n vertices if and only if $d_1 + \ldots + d_n$ is even and $\sum_{i=1}^{i=k} d_i \leq k(k-1) + \sum_{i=k+1}^{i=n} min(d_i, k)$, holds every $1 \leq k \leq n$.

Derangement

```
ll der[15];    der[0] = 1;    der[1] = 0;
REP(i,2,10) der[i] = ((ll)i-1)*(der[i-1]+der[i-2]);
```

Bell Number

The number of ways a set of n elements can be partitioned into nonempty subsets

```
ll bell[25+5][25+5];    bell[0][0] = 1;
REP(i,1,maks){
  bell[i][0] = bell[i-1][i-1];
  REP(j,1,i)bell[i][j]=bell[i-1][j-1]+bell[i][j-1];
} // Example : n=10 ==> bell[10][0]
```

Stirling Number 2nd Kind

The number of ways a set of n elements into k subsets)
Recurrence : S(n,k) = k*S(n-1,k) + S(n-1,k-1)

```
int stir[n+1][k+1];
for (int i=0; i<=n; i++) stir[i][0] = 0;
for (int i=0; i<=k; i++) stir[0][k] = 0;
REP(i,1,n) REP(j,1,i){
  if (j == 1 || i == j) stir[i][j] = 1;
  else stir[i][j] = j*stir[i-1][j] + stir[i-1][j↩
      -1];
} // Example : n=3,k=1 : stir[3][1]
```

Carmichael Number

Composite numbers n such that a^(n-1) == 1 (mod n) for every a coprime to n.

```
// 561, 1105, 1729, 2465, 2821, 6601, 8911, 10585
// 15841, 29341, 41041, 46657, 52633, 62745, 63973
// 75361, 101101, 115921, 126217, 162401, 172081
// 188461, 252601, 278545, 294409, 314821, 334153
// 340561, 399001, 410041, 449065, 488881, 512461
```

Vandermonde Convolution

$$\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k}, \qquad m,n,r \in \mathbb{N}_0,$$

Q-Vandermonde Convolution

$$\binom{n_1+\cdots+n_p}{m} = \sum_{k_1+\cdots+k_p=m} \binom{n_1}{k_1}\binom{n_2}{k_2}\cdots\binom{n_p}{k_p}$$

int m,n

$$\sum_k \binom{r}{m+k}\binom{s}{n-k} = \binom{r+s}{m+n}$$

int $l \geq 0$, int m,n

$$\sum_k \binom{l}{m+k}\binom{s}{n+k} = \binom{l+s}{l-m+n}$$

int $l \geq 0$, int m,n

$$\sum_k \binom{l}{m+k}\binom{s+k}{n}(-1)^k = (-1)^{l+m}\binom{s-m}{n-l}$$

int l,m, $m \geq 0$

$$\sum_{k\leq 1} \binom{l-k}{m}\binom{s}{k-n}(-1)^k = (-1)^{l+m}\binom{s-m-1}{l-m-n}$$

int $l, m \geq 0$, int $n \geq q \geq 0$

$$\sum_{0\leq k\leq l} \binom{l-k}{m}\binom{q+k}{n} = \binom{l+q+1}{m+n+1}$$

6.5 Necklace and Bracelets Combinatorics

$$N_k(n) = \frac{1}{n}\sum_{d|n}\varphi(d)k^{\frac{n}{d}}$$

Aperiodic Necklace: change φ to μ Mobius

$$B_k(n) = \begin{cases} \frac{1}{2}N_k(n) + \frac{1}{4}(k+1)k^{\frac{n}{2}} & \text{if } n \text{ is even} \\ \frac{1}{2}N_k(n) + \frac{1}{2}k^{\frac{n+1}{2}} & \text{if } n \text{ is odd} \end{cases}$$

6.6 Mathematics Sequence Formula

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6} \qquad \sum_{i=1}^{n} i^3 = \frac{n^2(n+1)^2}{4}$$

Triangular number sum

```
S(n) = ((n)(n+1)(n+2))/6
n = (sqrt(8x + 1) -1) / 2
```

$$\sum_{k=0}^{n} kCm = (n+1)C(m+1)$$

Faulhaber $O(p^2)$

$$\sum_{k=1}^{n} k^p = \frac{1}{p+1}\sum_{i=0}^{p}(-1)^i\binom{p+1}{i}B_i n^{p+1-i}$$

$$B_n = -\sum_{k=0}^{n-1}\binom{n}{k}\frac{B_k}{n-k+1}$$ for $n \geq 0$ and $B_n = 1$ for $n = 0$

6.7 Prime Number List

10^ 1	7	10^11	99999999977
10^ 2	97	10^12	999999999989
10^ 3	997	10^13	9999999999971
10^ 4	9973	10^14	99999999999973
10^ 5	99991	10^15	999999999999989
10^ 6	999983	10^16	9999999999999937

```
10^ 7   9999991       10^17   99999999999999997
10^ 8   99999989      10^18   999999999999999989
10^ 9   999999937     10^19   9999999999999999961
10^10   9999999967
Below (2^31)-1: 2147483629 (is 8th Mersenne Prime)
Below (2^32)-1: 4294967291
Below (2^63)-1: 9223372036854775783
Below (2^64)-1: 18446744073709551557
```

6.8 Catalan and Super Catalan

```
ll superCatalan(int n){
  if(n <= 2)return 1;
  return (3*(2*n-3)*sc(n-1)-(n-3)*sc(n-2)) / n;
} // 1,1,1,3,11,45, 197, 903, 4279, 20793, 103049
ll catalan(int n){ // nCk(2n,n) / (n+1)
  if(n <= 1)return 1;
  return c(n-1) * 2 * (2*n-1) / (n+1);
} // 1,1,2,5,14,42, 132, 429, 1430, 4862, 16796
```

6.9 Digit Counting Factorial

```
double memo[11000];
void start(){
  memo[1] = log10(1);
  REP(i,2,10000)memo[i]=memo[i-1]+(double)log10(i);
}
int solve(int n){ return (int)memo[n]+1; }
```

6.10 INT to ROMAN

```
const string R[] = {"M","CM","D","CD","C","XC", \
    "L","XL","X","IX","V","IV","I"};
const int N[] = { 1000, 900, 500, 400, 100, 90, \
    50, 40, 10, 9, 5, 4, 1};

string toRoman(int x){
  if(x == 0)return "0"; // Not decimal 0 !
  int n = sizeof(N)/sizeof(int); // 13
  string res = "";
  FOR(i,n) while(x >= N[i]) x-=N[i], res+=R[i];
  return res;
}
```

6.11 Linear Diophantine

```
// given a and b for equation ax + by = GCD(a,b)
// find the x and y and returns GCD(a,b)
int extEuclid(int a, int b, int& x, int& y){
  if(b == 0){ x=1;  y=0;  return a; }
  int res = ext_euclid(b,a%b,x,y);
  int x1 = y, y1 = x-(a/b)*y;  x=x1;  y=y1;
  return res;
}
// Is ax + by = c has integral solution?
bool linearDiop(int a,int b,int c, int& x,int& y){
  int d = extEuclid(a,b,x,y);
  if(c%d != 0)return false;
  x *= c/d;  y *= c/d;  return true;
}
```

6.12 Chinese Remainder Theorem

```
// Assume that f(0)=1 and 0^0=1.
// f(n) = (n%10)^f(n/10) for all n bigger than
// zero. Please calculate f(n)%m. (2 <= n, m <=
// 10^9, x^y means the y-th power of x).
ll inverse(ll a,ll m){ return mPow(a,phi(m)-1,m); }
ll CRT( vector<pair<LL,int>> data ) {
  ll M = 1;
  For(auto x: data)M *= x.se;
  ll ret = 0LL;
  For(auto x: data){
    ll Mi = M/x.se;
    ll temp = x.fi*Mi*inverse(Mi,x.se);
    ret += temp;  ret %= M;
  }
  return ret;
}
int nums[16], n;  const int LIM=1<<10;
ll brute(ll base, int pos, ll M){
  if(pos==n)return base%M;
  int j = n;  ll pj = 1;
  while(j-1 >= pos){
    ll k = 1;
    FOR(loop,pj){
      K *= nums[j-1];
      if(k >= LIM)return 0;
    }
    pj = k;  j--;
  }
  return mPow(base,pj,M);
}
ll solve(int,ll);
ll doit(int,int);
pair<ll,int> solve2(int pos, ll M){
  ll a = nums[pos];
  if(GCD(a,M) == 1){
    ll b = solve(pos+1,phi((int)M));
    return mp(mPow(a,b,M),M);
  }
  return mp(brute(a,pos+1,M),M);
}
ll solve(int pos, ll M){
  if(M == 1)return 0;
  if(pos == n)return 1;
  if(nums[pos] == (ll)1)return a%M;
  if(nums[pos] == (ll)0)return 0;
  ll m = M;  vector< pair<ll,int> > data;
  for(ll i=2;i*i<=m;i++){
    if(m%i!=0)continue;
    ll p=1;
    while(m%i==0)m/=i,p*=i;
    data.pb(solve2(pos,p));
  }
  if(m>1)data.pb(solve2(pos, m));
  return CRT(data);
}
void normalize(){
  nums[n] = 1;  int newN = n;
  FOR(i,n){
    if(nums[i]==1)newN = i, break;
    else if(nums[i]==0){
      int ctr0 = 0;
      while(nums[i+ctr0] == 0)ctr0++;
      if(ctr0%2 == 0) nums[i] = 1, newN = i;
      else if(i-1 >= 0)nums[i-1] = 1, newN = i-1;
      break;
  } }
  n = newN;  if(nums[n-1] != 1)nums[n++] = 1;
}
ll doit(int x, int m){
  n=0;
  while(x)nums[n++] = x%10, x /= 10;
  normalize();
  return solve(0,m);
}
int main(){
  int x, m;
  FOR(ncase,getint()){
    scanf("%d %d",&x,&m);
    printf("%d\n",(int)doit(x,m));
} }
```

```
//CRT Willson
int inv(int a, int m){
  int m0 = m, t, q, x0 = 0, x1 = 1;
  if(m == 1)return 0;
  while(a > 1){
    q = a/m;  t = m;  m = a%m;  a = t;
    t = x0;  x0 = x1 - q*x0;  x1 = t;
  }
  if(x1 < 0)x1 += m0; // Make x1 positive
  return x1;
}
// k is size of num[] and rem[]. Returns the
// smallest number x such that:
//     x % num[0] = rem[0]
//     x % num[1] = rem[1]
//     ..................
//     x % num[k-2] = rem[k-1]
```

```
// Assumption: Numbers in num[] are pairwise
// coprime (gcd for every pair is 1)
int findMinX(int num[], int rem[], int k){
  int prod = 1, res = 0;
  FOR(i,k) prod *= num[i]; // Compute prod of nums
  FOR(i,k){
    int pp = prod/num[i];
    res += rem[i]*inv(pp, num[i])*pp;
  }
  return res % prod;
}
int driver(){
  int num[] = {3, 4, 5}, rem[] = {2, 3, 1};
  int k = sizeof(num)/sizeof(num[0]);
  cout << "x is "<< findMinX(num, rem, k);
}
```

6.13 Euler Totient (Phi) Function

```
// count x < n that are coprime to n
int phi(int n){ // n * PHI( 1 - 1/pf )
  if(n == 1)return 0;
  int res = n;
  REP(i,2,n/i){
    if(n%i == 0)res -= res/i;
    while(n%i == 0)n /= i;
  }
  return res - ( n!=1 ? res/n : 0 );
}

vi sieve_phi(int n){ // sieve version
  vector<bool> is_prime(n+1,true);  vi ret(n+1);
  FOR(i,n)ret[i] = i;
  REP(i,2,n) if(is_prime[i]) REPP(j,i,n,i){
    ret[j] -= ret[j]/i, is_prime[j] = false;
  }
  return ret;
}
void sievephi(int n){
  phi[1] = 0;  for(LL i=2; i<n; i++)phi[i] = i;
  REPP((ll)i,2,n, (i==(ll)2)?1:(ll)2)
    if(phi[i] == i) for(LL j=i; j<=n; j+=i)
        phi[j] = phi[j]*(ll)(i-(ll)1)/(ll)i;
  REP((ll)i,2,n) for(ll j=1,_n=n/i; j<=_n; j++)
    ans[i*j] += j*phi[i];
  REP((ll)i,2,n) ans[i] += ans[i-1];
}
int phi(int n){
  if(n == 1)return 0;
  int res = n;
  REPP(i,2,n/i,(i==2?1:2)){
    if(n%i)continue;
    while(n>1 && n%i==0) n/=i;
    res = (res%i) ? res*(i-1)/i : res/i*(i-1);
  }
  if(n > 1)res = res/n*(n-1);
  return res;
}
```

6.14 Prime Test

```
const int MAX = (int)1e8;
const int HALF = (int)5e7;
const int SQRT = 5000;  const int SIZE = MAX/16+2;
char SITO[SIZE];
#define NTH(n) ((n)&7)
#define ISPRIME(n) (SITO[(n)>>4]&(1<<NTH((n)>>1)))

bool isPrime(int n){
  if(n == 2)return true;
  else if(!(n&1))return false;
  else return ISPRIME(n);
}
void init(){
  FILL(SITO,0xFF);
  SITO[0] = 0xFE;
  REP(i,1,SQRT-1) if(ISPRIME(i<<1))
    REPP(j,2*i*(i+1),HALF-1,2*i+1)
      SITO[j>>3] &= ~(1 << NTH(j));
}
```

6.15 Russian Peasant (BigMul with Modulo)

```
ll multiply(ll a, ll b, ll M, ll res = 0){
  for(; b; b>>=1, a=(a<<1)%M) (res+=(b&1)*a) %= M;
  return res;
} // don't forget to re-change the modulo
```

6.16 Pollard Rho

```
ll pollard_rho(ll n){
  for(ll d=n,int c=2; d==n; c++){
    ll x = 2, y = 2;
    do{ // Russian peasant
      x = (mulmod(x,x,n)+c) % n;
      y = (mulmod(y,y,n)+c) % n;
      y = (mulmod(y,y,n)+c) % n;
      d = gcd(abs(x-y),n);
      if(d>1)break;
    } while(1);
  }
  pollard_rho(d);  pollard_rho(n/d);
}
```

6.17 Miller Rabin (Probabilistic Prime Test)

```
// n-1 = (2^r)*d, a = rnd[2..n-2] as witness number
bool isProbablyPrime(ll a){
  a = fast(a,d,n); if(a == 1)return true;
  FOR(i,r){
    if(i)a = multiply(a,a,n);
    if(a == n-1)return true;
  }
  return false;
}
bool isPrime(ll n, int iteration, bool res = true){
  ll d = n-1, r = 0; while(d%2 == 0) d /= 2, r++;
  FOR(i,iteration) res &= isProbablyPrime(a);
  return res;
}
```

6.18 Generalized Lucas Theorem

```
// Special Lucas : (n,k) % p^x
// fctp[n] =  Product of the integers less than or
// equal to n that are not divisible by p
// Precompute fctp
ll p;
ll E(ll n, int m){
  ll tot = 0;
  while(n!=0)tot += n/m, n /= m;
  return tot;
}
ll funct(ll n, ll base){
  ll ans = fast(fctp[base],n/base,base);
  return (ans*fctp[n%base]) % base;
}
ll F(ll n, ll base){
  ll ans = 1;
  while(n!=0) ans=(ans*funct(n,base))%base, n/=p;
  return ans;
}
ll special_lucas(ll n, ll r, ll base){
  p = fprime(base);
  ll pow = E(n,p)-E(n-r,p)-E(r,p);
  ll TOP = fast(p,pow,base)*F(n,base) % base;
  ll BOT = F(r,base)*F(n-r,base) % base;
  return (TOP*fast(BOT,totien(base)-1,base))%base;
}
```

6.19 Sum of Divisors

```
// this function returns the sum of the divisors
// of n. i.e div(3) will return sum of the number
// of divisors up to 3.
ll div(int n){
  if(n==0)return 0;
  ll res = 0, i;
  for(i=2; i<=n/i; i++)res += n/i;
  i--;  res = (n*2)-(i*i)+(res*2);  return res;
}
```

6.20 Range Sieve

```
void sieve(int L, int U, vector<int>& primes){
  memset(isnp,0,sizeof(isnp));
  int nto = U-L+1;
  for(auto nP: primes){
    if(nP > U/nP)break;
    int fnum = (L/nP)*nP;
    if(fnum < L)fnum += nP;
    if(fnum == nP)fnum += nP;
    REPP(k,fnum-L,nto,nP)isnp[k] = 1;
  }
  if(L <= 1)isnp[1-L] = 1;
  if(L <= 0)isnp[0-L] = 1;
}
```

6.21 Mobius Function

```
int M[MAXN];   // M=mobius
int SPF[MAXN]; // SPF=smallest pf
bool B[MAXN];  vi prima; // B=is prime

void mobius(){
  prima.clear();  memset(B,true,sizeof(B));
  B[0] = B[1] = false;
  REP(i,2,MAXN-1){
    if(B[i])prima.pb(i), SPF[i] = i;
    FOR(auto j: prima){
      if(!(i*j<MAXN && j<=SPF[i]))break;
      B[i*j] = false;  SPF[i*j] = j;
  } }
  REP(i,1,MAXN-1){
    if(i == 1)M[i]=1;
    else if(SPF[i] == i)M[i] = -1;
    else if((i/SPF[i])%SPF[i] == 0)M[i] = 0;
    else M[i] = M[i/SPF[i]]*M[SPF[i]];
} }
```

6.22 Geometric Transformation

Cermin: $\begin{bmatrix}\cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta\end{bmatrix}$ Rotasi: $\begin{bmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{bmatrix}$

Dilatasi: $\begin{bmatrix}k & 0 \\ 0 & k\end{bmatrix}$ Gusuran/Sharing: $\begin{bmatrix}1 & x \\ y & 1\end{bmatrix}$

Regangan: $\begin{bmatrix}x & 0 \\ 0 & y\end{bmatrix}$

6.23 Trigonometric Equations

```
sin(3A)    = -4 Sin A + 3 Sin A
cos(3A)    = 4 Cos A  3 Cos A
sin(A/2)   = ((1  Cos A)/2)^0.5
cos(A/2)   = ((1 + Cos A)/2)^0.5
Tan A + Tan B + Tan C = Tan A Tan B Tan C ; A+B+C= PI

 2 Sin A Cos B    = Sin(A+B) + Sin(A-B)
 2 Cos A Sin B    = Sin(A+B) - Sin(A-B)
 2 Cos A Cos B    = Cos(A+B) + Cos(A-B)
-2 Sin A Sin B    = Cos(A+B) - Cos(A-B)
```

Rumus Sin Cos Segitiga

```
a/sin A = b/sin B = c/sin C
a^2 = b^2 + c^2 - 2 bc Cos A
2*L = (ab Sin C) = (bc Sin A) = (ac Sin B)
L = (c^2 * sin A * sin B) / (2 * sin C)
```

Heron's Formula

```
L = (s(s-a)(s-b)(s-c))^0.5     s = (a+b+c)/2
r = L/ s         R = abc / 4L
```

Luas n-gon

L = (n r r Sin(360/n))/2

Lingkaran Singung Segitiga

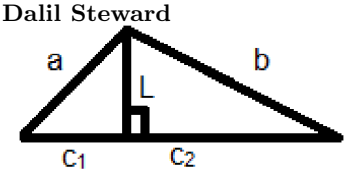
```
a: (s(s-b)(s-c)/(s-a))^0.5
b: (s(s-a)(s-c)/(s-b))^0.5
c: (s(s-a)(s-b)/(s-c))^0.5
```

$$m = \sqrt{\frac{2b^2 + 2c^2 - a^2}{4}}$$

Titik ke Garis

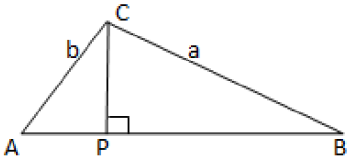
```
U = ((x3-x1)(x2-x1)+(y3-y1)(y2-y1))/((x2-x1)^2+(y2-y1)^2)
X = x1 + U(x2-x1)
Y = y1 + U(y2-y1)
```

Dalil Steward



$L^2.c = a^2.c_2 + b^2.c_1 - c_1.c_2.c$

Dalil Proyeksi



c

$AP = (b^2 + c^2 - a^2)/2c$

Law of Tangents

$$\frac{a-b}{a+b} = \frac{\tan^1/_2(\propto-\beta)}{\tan^1/_2(\propto+\beta)}$$

$$\frac{b-c}{b+c} = \frac{\tan^1/_2(\beta-\gamma)}{\tan^1/_2(\beta+\gamma)}$$

Mollweide's Formula

$$\frac{a+b}{c} = \frac{\cos^1/_2(\propto-\beta)}{\sin^1/_2\gamma}$$

$$\frac{a-c}{a+c} = \frac{\tan^1/_2(\propto-\gamma)}{\tan^1/_2(\propto+\gamma)}$$

7 Geometry

7.1 Polygon Centroid and Area

$$C_x = \frac{1}{6A}\sum_{i=0}^{n-1}(x_i + x_{i+1})(x_i\ y_{i+1} - x_{i+1}\ y_i)$$

$$C_y = \frac{1}{6A}\sum_{i=0}^{n-1}(y_i + y_{i+1})(x_i\ y_{i+1} - x_{i+1}\ y_i)$$

$$A = \frac{1}{2}\sum_{i=0}^{n-1}(x_i\ y_{i+1} - x_{i+1}\ y_i)$$

```
(x(n),y(n)) == (x0,y0)
```

7.2 Line Segment Intersection Point

$$Y = \frac{\begin{vmatrix} \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} & \begin{vmatrix} y_1 & 1 \\ y_2 & 1 \end{vmatrix} \\ \begin{vmatrix} x_3 & y_3 \\ x_4 & y_4 \end{vmatrix} & \begin{vmatrix} y_3 & 1 \\ y_4 & 1 \end{vmatrix} \end{vmatrix}}{\begin{vmatrix} \begin{vmatrix} x_1 & 1 \\ x_2 & 1 \end{vmatrix} & \begin{vmatrix} y_1 & 1 \\ y_2 & 1 \end{vmatrix} \\ \begin{vmatrix} x_3 & 1 \\ x_4 & 1 \end{vmatrix} & \begin{vmatrix} y_3 & 1 \\ y_4 & 1 \end{vmatrix} \end{vmatrix}} = \frac{\begin{vmatrix} \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} & y_1 - y_2 \\ \begin{vmatrix} x_3 & y_3 \\ x_4 & y_4 \end{vmatrix} & y_3 - y_4 \end{vmatrix}}{\begin{vmatrix} x_1 - x_2 & y_1 - y_2 \\ x_3 - x_4 & y_3 - y_4 \end{vmatrix}},$$

$$X = \frac{\begin{vmatrix} \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} & \begin{vmatrix} x_1 & 1 \\ x_2 & 1 \end{vmatrix} \\ \begin{vmatrix} x_3 & y_3 \\ x_4 & y_4 \end{vmatrix} & \begin{vmatrix} x_3 & 1 \\ x_4 & 1 \end{vmatrix} \end{vmatrix}}{\begin{vmatrix} \begin{vmatrix} x_1 & 1 \\ x_2 & 1 \end{vmatrix} & \begin{vmatrix} y_1 & 1 \\ y_2 & 1 \end{vmatrix} \\ \begin{vmatrix} x_3 & 1 \\ x_4 & 1 \end{vmatrix} & \begin{vmatrix} y_3 & 1 \\ y_4 & 1 \end{vmatrix} \end{vmatrix}} = \frac{\begin{vmatrix} \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} & x_1 - x_2 \\ \begin{vmatrix} x_3 & y_3 \\ x_4 & y_4 \end{vmatrix} & x_3 - x_4 \end{vmatrix}}{\begin{vmatrix} x_1 - x_2 & y_1 - y_2 \\ x_3 - x_4 & y_3 - y_4 \end{vmatrix}}$$

7.3 Line Segment Intersection

```
struct Vertex {int x, y;};
int GetArea2(Vertex& a, Vertex& b, Vertex& c) {
  return (b.x-a.x)*(c.y-a.y)-(c.x-a.x)*(b.y-a.y);
}
bool OnSegment(Vertex& a, Vertex& b, Vertex& c) {
  return(min(a.x, b.x) <= c.x && c.x <= max(a.x, b.↩
      x) && min(a.y, b.y) <= c.y && c.y <= max(a.y,↩
       b.y));
}
bool IsIntersect(Vertex& a, Vertex& b, Vertex& c, \
    Vertex& d){  int d1, d2, d3, d4;
  d1 = GetArea2(c, d, a);  d2 = GetArea2(c, d, b);
  d3 = GetArea2(a, b, c);  d4 = GetArea2(a, b, d);
  if(((d1 > 0 && d2 < 0) ||(d1 < 0 && d2 > 0)) &&((↩
       d3 > 0 && d4 < 0) || d3 < 0 && d4 > 0)){ ↩
      return true; }
  if(d1 == 0 && OnSegment(c, d, a)) {return true;}
  if(d2 == 0 && OnSegment(c, d, b)) {return true;}
  if(d3 == 0 && OnSegment(a, b, c)) {return true;}
  if(d4 == 0 && OnSegment(a, b, d)) {return true;}
  return false;
}
bool IsInside(Vertex& a, Vertex& lt, Vertex& rb) {
  return(min(lt.x, rb.x) <= a.x && a.x <= max(lt.x,↩
       rb.x) && min(lt.y, rb.y) <= a.y && a.y <= ↩
      max(lt.y, rb.y));
}
// line intersection
struct Vertex {
  int x, y; Vertex result;
  Vertex operator - (Vertex& b) {
    result.x = x - b.x;
    result.y = y - b.y;
    return result;
} };
void GetIntersectionAttribute(Vertex& a,Vertex& b,↩
    Vertex& c,Vertex& d,int& fN,int& sN,int& D){
  fN = (d.x-c.x)*(a.y-c.y) - (d.y-c.y)*(a.x-c.x);
  sN = (b.x-a.x)*(a.y-c.y) - (b.y-a.y)*(a.x-c.x);
  D = (d.y-c.y)*(b.x-a.x) - (d.x-c.x)*(b.y-a.y);
}
double GetIntersectionTime (int N, int D){
  return (double)N / (double)D;
}
double GetIntersectionX (Vertex& a, Vertex& b, ↩
    double t) {return a.x + (b - a).x * t;}
double GetIntersectionY (Vertex& a, Vertex& b, ↩
    double t) {return a.y + (b - a).y * t;}
```

7.4 Square Collision

```
((x1>=x1t && x1<=x2t) && ((y1>=y1t && y1<=y2t) ||
(y2>=y1t && y2<=y2t) || (y1<=y1t && y2>=y2t))) ||
((x2>=x1t && x2<=x2t) && ((y1>=y1t && y1<=y2t) ||
(y2>=y1t && y2<=y2t) || (y1<=y1t && y2>=y2t))) ||
((x1<=x1t && x2>=x2t) && ((y1>=y1t && y1<=y2t) ||
(y2>=y1t && y2<=y2t) || (y1<=y1t && y2>=y2t)))
```

7.5 Closest Pair

```
const double inf = 10000.0;
typedef long long ll;
struct point{ double x, y;  };
int cmp(const point &a, const point &b) {
  return a.x < b.x;
}
int n, buff[10005], Y[10005], Yak[10005];
point X[10005];
double conquer(int l, int mid, int r, double dn){
  for(int i=l,j=mid+1,k=l; i<=mid || j<=r; k++){
    if((j>r) || (i<=mid && X[Y[i]].y<X[Y[j]].y))
      buff[k] = Y[i], i++;
    else buff[k] = Y[j], j++;
  }
  int nak = 0;
  REP(i,l,r){
    Y[i] = buff[i];
    if(fabs(X[mid].x - X[Y[i]].x) <= dn){
      Yak[nak++] = Y[i];
  } }
  double ret = dn;
  FOR(i,nak) REP(j,i+1,nak-1){
    if (fabs(X[Yak[i]].y-X[Yak[j]].y)>ret) break;
    ret = min(ret,dist(X[Yak[i]], X[Yak[j]]));
  }
  return ret;
}
double divide(int l, int r){
  double ret = inf;
  if(l < r){
    double dl = divide(l, (l+r)/2);
    double dr = divide((l+r)/2+1, r);
    ret = conquer(l, (l+r)/2, r, min(dl, dr));
  }
  return ret;
}
int main(){
  while(scanf("%d", &n) != EOF && n){
    FOR(i,n) scanf("%lf %lf", &X[i].x, &X[i].y);
    sort(X,X+n, cmp);
    FOR(i,n) Y[i] = i;
    double ans = divide(0,n-1);
    if (fabs(ans-inf) <= EPS) printf("INF\n");
    else printf("%.4lf\n", ans);
} }
```

7.6 Polygon Class

```
class Line{
  public :
  Point p1, p2;
  Line(){}
  Line(Point _p1, Point _p2):p1(_p1),p2(_p2){}
  bool intersect(double x1, double y1, double x2, ↩
      double y2, double x3, double y3, double x4, ↩
      double y4,double &x, double &y) {
    double nom = (x2-x1)*(y1-y3) + (y2-y1)*(x3-x1);
    double den = (x2-x1)*(y4-y3)-(x4-x3)*(y2-y1);
    if (den==0) return false;
    double b = nom/den;
```

```
    x = x3 + (x4-x3) * b;
    y = y3 + (y4-y3) * b;
    double a = ((x-x1) / (x2-x1));
    if(x2-x1 == 0) a = ((y-y1) / (y2-y1));
    return 0<=a && a<=1 && 0<=b && b<=1;
  }
  bool intersect(Line &that, Point &p) {
    return intersect(p1.x, p1.y, p2.x, p2.y, \
      that.p1.x, that.p1.y, that.p2.x, that.p2.y, \
      p.x, p.y);
  }
};
class Polygon {
  public :
  VP P;
  Polygon(){}
  Polygon(VP _P):P(_P){}
  void addPoint(Point p){P.push_back(p);}
  Polygon intersect(Polygon &that) {
    double qarea = that.area(), parea = area();
    VP &Q = that.P;
    set<Point> p;
    Point pt;
    REP(i,P.size()){
      int ni = (i+1)%P.size();
      Point p1(P[i].x, P[i].y);
      Point p2(P[ni].x,P[ni].y);
      Line L1(p1, p2);
      REP(j,Q.size()){
        int nj = (j+1)%Q.size();
        Point p3(Q[j].x, Q[j].y);
        Point p4(Q[nj].x,Q[nj].y);
        Line L2(p3, p4);
        if (L1.intersect(L2,pt)) p.insert(pt);
    } }
    REP(i,P.size()){
      double a = 0;
      REP(j,Q.size()) a += fabs(det(Q[j],
      Q[(j+1)%Q.size()], P[i])/2);
      if (fabs(a-qarea) < EPS) p.insert(P[i]);
    }
    REP(i,Q.size()){
      double a = 0;
      REP(j,P.size()) a += fabs(det(P[j],
      P[(j+1)%P.size()], Q[i])/2);
      if (fabs(a-parea) < EPS) p.insert(Q[i]);
    }
    return Polygon(cHull(VP(p.begin(),p.end())));
  }
  double area(){
    double ret = 0;
    REP(i, P.size()){
      int ni = (i+1)%P.size();
      ret += P[i].x * P[ni].y - P[i].y * P[ni].x;
    }
    return fabs(ret/2);
} };
int main(){
  for(int nA,nB; scanf("%d",&nA)!=EOF && nA; ){
    Polygon P, Q;
    FOR(i,nA){
      int x,y; scanf("%d %d",&x,&y);
      P.addPoint(Point(x,y));
    }
    scanf("%d",&nB);
    FOR(i,nB){
      int x,y; scanf("%d %d",&x,&y);
      Q.addPoint(Point(x,y));
    }
    Polygon I = P.intersect(Q);
    printf("%lf\n",P.area()+Q.area()-2*I.area());
  }
  puts("");
  return 0;
}
```

7.7 Point to Line

```
// Returns the distance from p to the line defined
// by {a, b}.The closest point on the line is
// returned through (cpx,cpy). Does not work for
// degenerate lines (when answer isundefined).
// REQUIRES: dist()
// Formula: cp = a + (p-a).(b-a) / |b-a| * (b-a)
double distToLine(
  double ax, ay, bx, by, px, py;
  double *cpx, double *cpy ) {
    double a = ((px-ax)*(bx-ax) + (py-ay)*(by-ay));
    double b = ((bx-ax)*(bx-ax) + (by-ay)*(by-ay));
    double proj = a/b;
    *cpx = ax + proj * ( bx - ax );
    *cpy = ay + proj * ( by - ay );
    return dist(px, py, *cpx, *cpy);
}
```

7.8 Line to Segment

```
// Returns the distance from p to the line segment
// ab. Theclosest point on ab is returned through
// (cpx, cpy). Workscorrectly for degenerate line
// segments (a == b).
// REQUIRES: dist(), distToLine()
double distToLineSegment(
  double ax, double ay, double bx, double by,
  double px, double py, double *cpx, double *cpy){
    if((bx-ax)*(px-ax) + (by-ay)*(py-ay) < EPS){
      *cpx = ax;  *cpy = ay;
      return dist(ax, ay, px, py);
    }
    if((ax-bx)*(px-bx) + (ay-by)*(py-by) < EPS){
      *cpx = bx;  *cpy = by;
      return dist(bx, by, px, py);
    }
    return distToLine(ax,ay,bx,by, px,py,cpx,cpy);
}
```

7.9 Polygon Diameter

```
struct ConvexPolygon {
  Point *p;  int n;
  ConvexPolygon(Point* pp, int nn){ p=pp; n=nn; }
double diameter() {
  int a, b;  double d=(double)0.0, area1, area2;
  if(n == 2) return getDist(p[0], p[1]);
  if(n == 3) return max(getDist(p[0],p[1]), \
    getDist(p[1], p[2]), getDist(p[0], p[2]));
  a = n-1; b = 1;
  area1 = area(p[a], p[(a+1)%n], p[(b+1)%n]);
  area2 = area(p[a], p[(a+1)%n], p[b]);
  while(area1 > area2){
    b = (b+1) % n;
    area1 = area(p[a], p[(a+1)%n], p[(b+1)%n]);
    area2 = area(p[a], p[(a+1)%n], p[b]);
  }
  while(b != 0){
    a = (a+1)%n;
    d = max(d, distance2Point(p[a], p[b]));
    area1 = area(p[a], p[(a+1)%n], p[(b+1)%n]);
    area2 = area(p[a], p[(a+1)%n], p[b]);
    while(area1 > area2){
      b = (b+1)%n;
      area1 = area(p[a], p[(a+1)%n], p[(b+1)%n]);
      area2 = area(p[a], p[(a+1)%n], p[b]);
      if(b == 0) return d;
      d = max(d, distance2Point(p[a], p[b]));
  } }
  return d;
} };
```

7.10 Union Rectangle Intersection

```
struct event{ int ind, type;};
struct point{ double x,y; };

event events_v[350],events_h[350];
point rect[150][3];
bool in_set[350];
int n;

bool compare_x(event a,event b){ return rect[a.ind↩
    ][a.type].x<rect[b.ind][b.type].x; }
bool compare_y(event a,event b){ return rect[a.ind↩
    ][a.type].y<rect[b.ind][b.type].y; }
```

```
double union_area(int e){
  memset(in_set,false,sizeof(in_set));
  sort(events_v,events_v + e, compare_x);
  sort(events_h,events_h + e, compare_y);
  double area = 0;
  in_set[events_v[0].ind] = 1;
  REP(i,1,e-1){
    event c = events_v[i];
    int cnt = 0;
    double delta_x = rect[c.ind][c.type].x - rect[↩
        events_v[i-1].ind][events_v[i-1].type].x;
    double begin_y;
    if(delta_x == 0){
      in_set[c.ind] = (c.type == 0);  continue;
    }
    FOR(j,e){
      if(in_set[events_h[j].ind] == 1){
        if(events_h[j].type == 0){
          if(cnt == 0)
            begin_y = rect[events_h[j].ind][0].y;
          ++cnt;
        }
        else if(--cnt == 0){
          double delta_y = \
            rect[events_h[j].ind][1].y - begin_y;
          area += delta_x * delta_y;
    } } }
    in_set[c.ind ] = (c.type == 0);
  }
  return area;
}
int main(){
  int tc = 0;
  while(cin>>n && n!=0){
    tc++;
    FOR(a,n){
      double x,y,r;  cin>>x>>y>>r;
      rect[a][0].x = x-r;
      rect[a][0].y = y-r;
      rect[a][1].x = x+r;
      rect[a][1].y = y+r;
      events_v[2*a].ind = a;
      events_v[2*a].type = 0;
      events_v[2*a + 1].ind = a;
      events_v[2*a + 1].type = 1;
      events_h[2*a].ind = a;
      events_h[2*a].type = 0;
      events_h[2*a + 1].ind = a;
      events_h[2*a + 1].type = 1;
    }
    double area = union_area(2*n);
    printf("%d %.2lf\n",tc,area);
} }
```

7.11 Great Circle Distance

```
const double R = 6371.01;
double a=sin(plat)*sin(qlat);
double b=cos(plat)*cos(qlat)*cos(plong)*cos(qlong);
double c=cos(plat)*cos(qlat)*sin(plong)*sin(qlong);
return R * acos(a + b + c);
```

8 Geometric Paper

